

Can LLM-Assisted Control Parameter Design Outperform Domain Experts in Dynamic Launch Vehicle Attitude Control?

Yongbo Ma¹, Cong Huang², Weiwei Yuan¹, Peng Jiafeng¹, Shijing Guo¹, Yuhao Zhang¹, Wenwen Zheng¹, Donghai Guan¹, and Hongsong Peng¹

¹College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics
Nanjing, China

²Beijing Institute of Aerospace Automatic Control, China Academy of Launch Vehicle Technology
Beijing, China

Abstract

Dynamic Launch Vehicle Attitude Control (DLVAC) is critical to flight safety and can be formulated as a dynamic optimization problem to design control parameters. Existing methods primarily rely on either experienced domain experts or optimization algorithms. However, the distinctive mathematical characteristics of control parameters lead to variable structures with varying dimensionality. This variability reshapes the search space, limiting both experts and algorithms in exploring promising structures and thereby degrading control performance. This paper presents **LLM-Ctrl**, a novel framework employing Large Language Models (LLMs) as assistant agents for control parameter design. Specifically, the framework introduces a new LLM-assisted paradigm via two complementary mechanisms: (1) a coevolution activation mechanism that enables variable structure evolution, where an LLM agent triggers the exploration of diverse control parameter structures within flight stages; and (2) a search reinitialization mechanism in which another LLM agent reasons over historical design trajectories to provide promising structures across flight stages. Extensive experiments on five DLVAC tasks demonstrate that, within the same time budget, LLMCtrl outperforms domain experts and improves the attitude control success rate by up to 12.56%. Furthermore, LLM-Ctrl consistently outperforms state-of-the-art algorithms that we adapted for DLVAC across all tasks. To our knowledge, we provide the first publicly available data-driven DLVAC tasks and the code at <https://github.com/LLMCtrl/LLMCtrl>.

CCS Concepts

• Computing methodologies → Search methodologies; • Applied computing → Aerospace.

Keywords

Dynamic Launch Vehicle Attitude Control, LLM-Assisted Control Parameter Design, Dynamic Optimization Algorithm

ACM Reference Format:

Yongbo Ma¹, Cong Huang², Weiwei Yuan¹, Peng Jiafeng¹, Shijing Guo¹, Yuhao Zhang¹, Wenwen Zheng¹, Donghai Guan¹, and Hongsong Peng¹. 2018. Can LLM-Assisted Control Parameter Design Outperform Domain Experts in Dynamic Launch Vehicle Attitude Control?. In *Proceedings of Conference Full Name (Conference acronym 'XX)*, June 03–05, 2018, Woodstock, NY. ACM, New York, NY, USA, 15 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Dynamic Launch Vehicle Attitude Control (DLVAC) aims to ensure the safety and stability of launch vehicle flight under complex conditions [2, 26, 41, 46]. As illustrated in Figure 1, a launch vehicle flight process consists of multiple continuous flight stages, each represented by a set of transfer functions. In practice, transfer functions model flight states offline while capturing extreme deviations, the designed control parameters are executed online by onboard hardware. DLVAC can therefore be formulated as a control parameter design task, in which each stage requires a variable-structure solution to regulate all transfer functions simultaneously.

Motivations. DLVAC seeks optimal control parameters across changing flight stages, where structural variations continuously reshape the search space. From a structural perspective, its variability arises from the uncertain dimensionality of control components. Transfer functions can accommodate control parameters with diverse structures, and control performance depends on both structure and parameter values. However, the vast structure space exceeds the design capability of both domain experts and existing optimization algorithms. From a dynamic perspective, designing control parameters across different flight stages requires extensive domain knowledge. Currently, highly experienced domain experts can potentially design reliable control parameters for the entire flight process, but through labor-intensive manual design. Meanwhile, dynamic optimization algorithms [11, 19, 27, 28, 31] struggle to balance exploration and exploitation as structural variations continually reshape the decision space.

Domain expertise demands substantial time and resources, and existing optimization algorithms lack the domain knowledge required for effective control parameter design. Recent advances in Large Language Models (LLMs) [1, 17, 22, 32, 36] have demonstrated their strong reasoning capabilities across complex interdisciplinary tasks, motivating us to introduce an LLM-assisted paradigm for control parameter design in launch vehicle attitude control.

Challenges. Despite their potential to rival domain experts through advanced generalization and reasoning capabilities, LLMs cannot be directly applied to control parameter design without

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

Conference acronym 'XX, Woodstock, NY

© 2018 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-XXXX-X/2018/06

<https://doi.org/XXXXXXX.XXXXXXX>

dedicated mechanisms. The core challenges can be summarized into two key aspects:

(1) **Structural Variations of Control Parameters.** The control parameter structure is determined by the number and configuration of zeros and poles in the transfer function obtained through the Laplace transform. Therefore, control parameters with different structures can achieve attitude control, and the key challenge is to identify the optimal structure and its corresponding parameter values. Although optimization algorithms can search beyond domain experts' structural preferences and explore diverse structures, the variations continually reshape the decision space, rendering the provided optimization directions unreliable.

(2) **Latent Design Knowledge Across Stages.** Different flight stages contain informative and exploitable design knowledge. The key challenge lies in identifying what knowledge is transferable and designing how to exploit it across stages. Heterogeneous structural preferences of domain experts hinder the abstraction of individual experience into unified design knowledge. Existing optimization algorithms [8, 48] remain inadequate for DLVAC because the coupled changes in parameter structure and flight stages undermine their ability to reuse historical search experience.

Contributions. To this end, we propose LLMCtrl, a novel framework that establishes an LLM-assisted paradigm in which two complementary mechanisms jointly design control parameter structures for DLVAC. Our main contributions are summarized as follows:

(1) **A New LLM-Assisted Paradigm.** We pioneer the use of advanced LLMs as assistant agents that integrate domain knowledge and perform reasoning with explicit consideration of DLVAC characteristics. To the best of our knowledge, LLMCtrl represents the first attempt to incorporate LLM-assisted expert-like reasoning into DLVAC for control parameter design.

(2) **Two Complementary Mechanisms.** The coevolution activation mechanism employs the Designer Agent for within-stage exploration of diverse control parameter structures, while the search reinitialization mechanism uses the Advisor Agent for cross-stage reasoning over historical design trajectories to identify promising structures for new stages.

(3) **Outperform Domain Experts on five DLVAC tasks.** LLMCtrl outperforms domain experts and improves the attitude control success rate by up to 12.56% across the DLVAC tasks under the same time budget. Moreover, we develop a unified approach for adapting existing algorithms to DLVAC, and LLMCtrl consistently outperforms the resulting state-of-the-art algorithms.

2 Dynamic Launch Vehicle Attitude Control

2.1 Dynamic Flight Process

The dynamic flight process $\{\mathcal{E}^t\}_{t=1}^T$ involves a sequence of T flight stages. In each stage $\mathcal{E}^t$, the launch vehicle state is represented by a set of transfer functions:

$$\mathcal{G}^t = \{G_k^t(s)\}_{k=1}^{K_t}, \quad (1)$$

where s represents the complex frequency-domain variable, K_t is the number of transfer functions in stage t . Both K_t and the settings of $G_k^t(s)$ may vary over time, for the k -th transfer function in stage

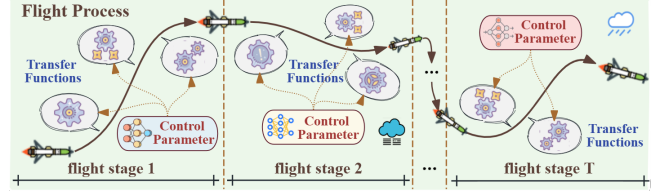


Figure 1: Dynamic Launch Vehicle Attitude Control.

t , its general form is defined as

$$G_k^t(s) = \kappa_k^t \cdot \frac{\prod_{a=1}^{N_{1,k}^t} (O_{a,k}^t s + 1) \prod_{b=1}^{N_{2,k}^t} \left[\left(\frac{s}{\omega_{b,k}^t} \right)^2 + \frac{2\xi_{b,k}^t}{\omega_{b,k}^t} s + 1 \right]}{\prod_{c=1}^{N_{3,k}^t} (O_{c,k}^t s + 1) \prod_{d=1}^{N_{4,k}^t} \left[\left(\frac{s}{\omega_{d,k}^t} \right)^2 + \frac{2\xi_{d,k}^t}{\omega_{d,k}^t} s + 1 \right]}, \quad (2)$$

where κ defines the gain of the transfer function, $N_{1,k}^t, N_{2,k}^t, N_{3,k}^t$, and $N_{4,k}^t$ represent the numbers of first-order numerator terms, second-order numerator terms, first-order denominator terms, and second-order denominator terms, respectively. $O_{a,k}^t$ and $O_{c,k}^t$ are the time variation, $(\omega_{b,k}, \omega_{d,k})$ and $(\xi_{b,k}, \xi_{d,k})$ are the frequency and damping ratios of the second-order differential and oscillation element.

2.2 Variable-Structure Control Parameter

The structure of the control parameter $\mathbf{x}^t$ at the t -th stage follows the identical form of the transfer functions. However, the control parameter can adopt variable-structures, let N_1, N_2, N_3 , and N_4 denote the number of numerator and denominator terms in $G_k^t(s)$, the structure of $\mathbf{x}^t$ is defined as

$$D_{\mathbf{x}^t} = N_1 + 2N_2 + N_3 + 2N_4, \quad (3)$$

where each second-order term contributes two parameters, and the structural orders are bounded by domain expert knowledge as $1 \leq N_r \leq N_{\text{exp}}, r \in \{1, 2, 3, 4\}$.

2.3 Task Formulation

For the t -th stage, DLVAC searches for a control parameter $\mathbf{x}^t \in \mathbb{R}^{D_{\mathbf{x}^t}}$ with a variable-structure to regulate all the transfer functions in $\mathcal{G}^t$. Based on the Bode plots of $\mathcal{G}^t$, we extract the flight attitude indicators and formulate them as multiple objectives to be optimized. The detailed rules for indicators are presented in **Appendix A**. The unified maximization objective vector for the k -th transfer function $G_k^t(s)$ is defined as

$$\mathbf{f}_k^t(\mathbf{x}^t) = \left[\omega_{0,k}^t, A_{d,k}^t, -A_{g,k}^t, -A_{h,k}^t, -A_{e,k}^t, \varphi_{d,k}^t, \varphi_{h,k}^t, \varphi_{e,k}^t \right]^\top. \quad (4)$$

We define the objectives for each stage as the worst objective value over the transfer function set, each of the eight objectives is aggregated at the stage level as

$$F_j^t(\mathbf{x}^t) = \min_{1 \leq k \leq K_t} f_{j,k}^t(\mathbf{x}^t), \quad (5)$$

where $f_{j,k}^t(\mathbf{x}^t)$ denotes the j -th component of $\mathbf{f}_k^t(\mathbf{x}^t)$ and the aggregation guides the control parameter to improve the most difficult

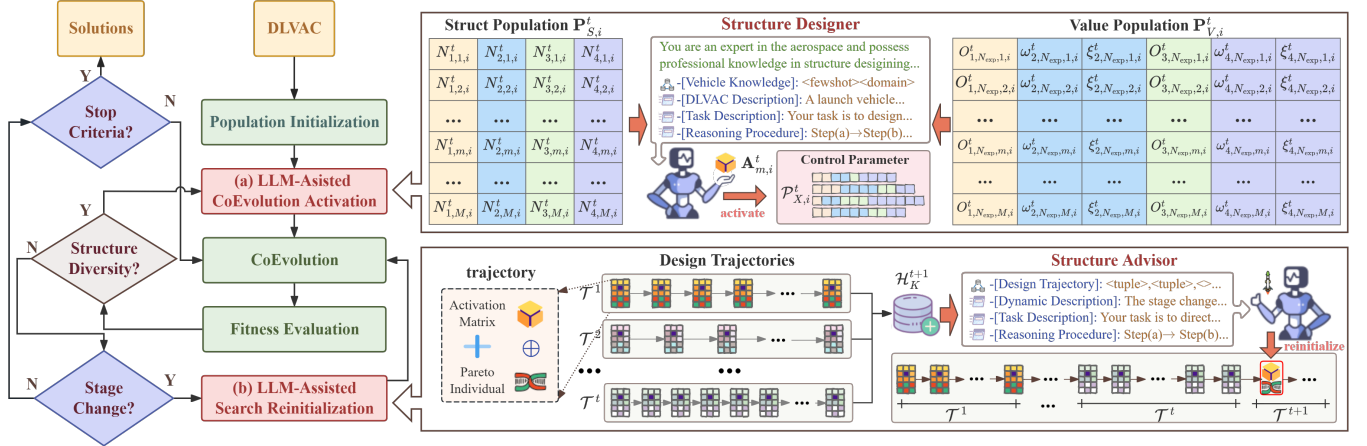


Figure 2: Overview of LLMCtrl. LLM-Assisted Coevolution Activation explores variable-structure control parameters, and LLM-Assisted Search Reinitialization exploits design trajectories to guide structural search in subsequent flight stages.

transfer function for each objective. The DLVAC task is then formulated to search for a sequence of control parameters $\{\mathbf{x}^t\}_{t=1}^T$:

$$\max_{\mathbf{x}^t \in \mathbb{R}^{D_{\mathbf{x}^t}}} \{\mathcal{F}^t(\mathbf{x}^t)\}_{t=1}^T = \{F_1^t(\mathbf{x}^t), F_2^t(\mathbf{x}^t), \dots, F_8^t(\mathbf{x}^t)\}_{t=1}^T. \quad (6)$$

We highlight two key characteristics of DLVAC task: (i) the control parameter $\mathbf{x}^t$ as the decision vector, has a dimension $D_{\mathbf{x}^t}$ that varies with its parameter structure; (ii) each objective $F_j^t(\mathbf{x}^t)$ is constructed by aggregating the same flight attitude indicators across multiple transfer functions.

3 LLMCtrl for Control Parameter Design

We introduce two key mechanisms of LLMCtrl: LLM-Assisted Coevolution Activation and LLM-Assisted Search Reinitialization. Integrated into the dynamic optimization process, the two mechanisms coordinate LLM agents to explore variable-structure control parameters and leverage historical design trajectories for subsequent flight stages. Figure 2 provides the overview of our framework. The detailed pseudocode is described in **Appendix B**.

3.1 LLM-Assisted Coevolution Activation

We construct a structure population and a value population with the same population size M for each stage t undergoes I_t iterations. By activating these populations, the variable-structure control parameters efficiently participate in the coevolution.

3.1.1 Coevolution Populations. The structure population $\mathbf{P}_{S,i}^t$ and value population $\mathbf{P}_{V,i}^t$ at iteration i in stage t are defined as

$$\begin{aligned} \mathbf{P}_{S,i}^t &= [\mathbf{n}_{1,i}^t, \dots, \mathbf{n}_{m,i}^t, \dots, \mathbf{n}_{M,i}^t]^\top \in \mathbb{Z}_{>0}^{M \times 4}, \\ \mathbf{P}_{V,i}^t &= [\mathbf{P}_{V1,i}^t, \mathbf{P}_{V2,i}^t, \mathbf{P}_{V3,i}^t, \mathbf{P}_{V4,i}^t] \in \mathbb{R}^{M \times 6N_{\text{exp}}}, \end{aligned} \quad (7)$$

where $\mathbf{n}_{m,i}^t = [N_{1,m,i}^t, N_{2,m,i}^t, N_{3,m,i}^t, N_{4,m,i}^t]^\top$ defines the structure of the control parameter by specifying the number of terms, each term takes an integer value from $\{1, \dots, N_{\text{exp}}\}$. The value population is partitioned into four blocks, where $\mathbf{P}_{V1,i}^t$ and $\mathbf{P}_{V3,i}^t$ optimize the first-order value for each individual, $\mathbf{P}_{V2,i}^t$ and $\mathbf{P}_{V4,i}^t$ optimize the second-order values. The detailed definition is provided in **Appendix C**.

3.1.2 Structure Designer. We employ the LLM agent to identify the candidate positions to be activated in $\mathbf{P}_{V,i}^t$ based on the numbers in $\mathbf{P}_{S,i}^t$. The agent determines the activation pattern by generating an activation matrix using Chain-of-Thought (CoT) [9, 45] reasoning.

Domain-Guided Structure Reasoning. We provide the Designer with the comprehensive DLVAC information, which includes the launch vehicle knowledge $\mathcal{K}_V$, the DLVAC description $\mathcal{D}_{\text{DLVAC}}$, and the task description $\mathcal{D}_{\text{task}}$. The Designer can therefore better understand the task, leading to reasonable activation and reliable outputs. Formally, we denote the composed prompt as $Q_{D,i}^t = \text{Prompt}(\mathcal{K}_V, \mathcal{D}_{\text{DLVAC}}, \mathcal{D}_{\text{task}}, \mathbf{P}_{S,i}^t, \mathbf{P}_{V,i}^t)$. The detailed prompt template is shown in **Appendix D**.

We then instruct the agent to perform reasoning using the constructed prompt. The agent first establishes the relationship between attitude control and its role in designing the control parameter structure. It subsequently generates a rank matrix shared by the population, which encodes the importance of the structure term:

$$\mathbf{R}_i^t = \Pi_{\text{Designer}}(Q_{D,i}^t) = [\mathbf{r}_{1,i}^t, \mathbf{r}_{2,i}^t, \mathbf{r}_{3,i}^t, \mathbf{r}_{4,i}^t]^\top \in \mathbb{R}^{4 \times N_{\text{exp}}}, \quad (8)$$

where $\mathbf{r}_{j,i}^t \in \mathbb{R}^{N_{\text{exp}}}$ denotes the rank score vector generated for the j -th structure term, and each element specifies the priority of a possible position related to $\mathbf{P}_{V,i}^t$.

Activation Matrix. For the m -th structure individual, the activation matrix is constructed from $\mathbf{R}_i^t$ according to its value:

$$\mathbf{A}_{m,i}^t = [\mathbf{a}_{1,m,i}^t, \mathbf{a}_{2,m,i}^t, \mathbf{a}_{3,m,i}^t, \mathbf{a}_{4,m,i}^t]^\top \in \{0, 1\}^{4 \times N_{\text{exp}}}, \quad (9)$$

where $\mathbf{a}_{r,m,i}^t \in \{0, 1\}^{N_{\text{exp}}}$ denotes the activation vector for the r -th structure term of the m -th individual. Its q -th element is defined as

$$a_{r,q,m,i}^t = \mathbb{I}[q \in \text{TopK}(\mathbf{r}_{r,i}^t, N_{r,m,i}^t)],$$

where q indexes the N_{exp} candidate positions, $\text{TopK}(\mathbf{r}, N)$ returns the indices corresponding to the N largest entries of $\mathbf{r}$, and $\mathbb{I}[\cdot]$ denotes the indicator function. This construction activates exactly $N_{r,m,i}^t$ positions and thus satisfies $\sum_{q=1}^{N_{\text{exp}}} a_{r,q,m,i}^t = N_{r,m,i}^t$.

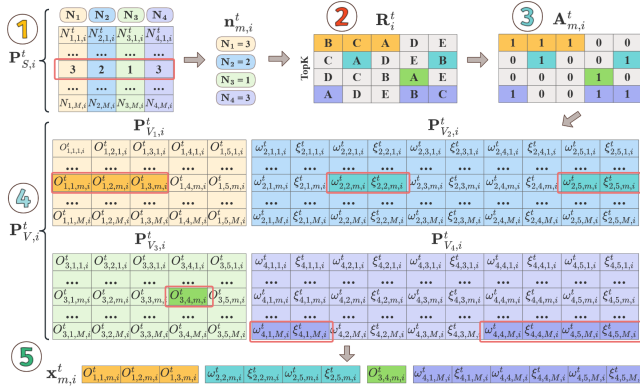


Figure 3: Illustration of LLM-Assisted Coevolution Activation. $\mathbf{P}_{S,i}^t$ defines the number of structure terms, $\mathbf{R}_i^t$ specify the activation pattern $\mathbf{A}_{m,i}^t$. The activated values in $\mathbf{P}_{V,i}^t$ are then constructed as the variable-structure control parameter $\mathbf{x}_{m,i}^t$.

3.1.3 Variable Structure Population. As illustrated in Figure 3, for each structure individual $\mathbf{n}_{m,i}^t$ in $\mathbf{P}_{S,i}^t$, an activation matrix $\mathbf{A}_{m,i}^t$ is derived from $\mathbf{R}_i^t$. Specifically, $\mathbf{n}_{m,i}^t$ determines the number of activated positions and $\mathbf{A}_{m,i}^t$ specifies the activation pattern. Then, the corresponding values in $\mathbf{P}_{V,i}^t$ are subsequently activated and constructed into a variable-structure control parameter $\mathbf{x}_{m,i}^t$. Finally, the variable-structure population can be expressed as

$$\mathcal{P}_{X,i}^t = \left\{ \mathbf{x}_{m,i}^t = \text{Construct} \left(\mathbf{n}_{m,i}^t, \mathbf{P}_{V,i}^t; \mathbf{A}_{m,i}^t \right) \right\}_{m=1}^M. \quad (10)$$

3.1.4 Structure Designer Interaction. To avoid excessive interference, we invoke the Structure Designer only when $\mathbf{P}_{S,i}^t$ exhibits insufficient diversity. For two structure individuals $\mathbf{n}_{m,i}^t$ and $\mathbf{n}_{n,i}^t$, their normalized distance is defined as $d_{m,n,i}^t = \|\mathbf{n}_{m,i}^t - \mathbf{n}_{n,i}^t\| / [4(N_{\text{exp}} - 1)]$, the denominator represents the maximum possible difference across the four structure terms. The structural diversity is defined as

$$d_{S,i}^t = \frac{2}{M(M-1)} \sum_{m=1}^{M-1} \sum_{n=m+1}^M d_{m,n,i}^t. \quad (11)$$

A smaller $d_{S,i}^t$ indicates stronger structural similarity and insufficient diversity. When $d_{S,i}^t$ falls below a predefined threshold $\epsilon_S \in [0, 1]$, the Structure Designer generates a new rank matrix $\mathbf{R}_i^t$.

3.2 LLM-Assisted Search Reinitialization

Although coevolution activation explores variable structures, stage transitions may invalidate previously optimized designs. The Structure Advisor therefore leverages historical design trajectories to reinitialize structural search for the next flight stage.

3.2.1 Design Trajectory. Let C^t denote the set of iteration indices at which the Designer is invoked in stage t . After each invocation $i \in C^t$, we identify the nondominated control parameters in $\mathcal{P}_{X,i}^t$, and their individual indices are defined as

$$\mathcal{I}_{\mathcal{P},i}^t = \left\{ m \mid \mathbf{x}_{m,i}^t \in \text{ND}(\mathcal{P}_{X,i}^t) \right\}, \quad (12)$$

where $\text{ND}(\cdot)$ returns the nondominated solutions, and each of the structure individual is paired with the activation matrix. Therefore,

Table 1: Details of the DLVAC tasks with target thresholds.

Objectives	Task-1	Task-2	Task-3	Task-4	Task-5
Cut-Off Freq ω_0	[0.1,10]	[0.01,5]	[0.01,15]	[2,9]	[0.5,10]
Low-Freq Amp A_d	4.0	4.0	7.4	4.5	8.0
High-Freq Amp A_g	-8.0	-5.5	-11.0	-4.0	-7.5
Elastic Amp A_e	—	-5.0	-6.3	-9.5	-5.0
Oscillation Amp A_h	—	—	—	-10.0	-7.8
Low-Freq Phase φ_d	30.0	16.0	22.0	18.6	14.4
Oscillation Phase φ_h	—	26.0	30.0	—	—
Elastic Phase φ_e	—	—	16.0	—	24.3

the design trajectory is defined as

$$\tau_i^t = \left\{ (\mathbf{A}_{m,i}^t, \mathbf{n}_{m,i}^t) \mid m \in \mathcal{I}_{\mathcal{P},i}^t \right\}, \quad (13)$$

where $\mathbf{A}_{m,i}^t$ records the selected structure terms and $\mathbf{n}_{m,i}^t$ specifies the number of structure terms. All design trajectories in stage t form the repository $\mathcal{T}^t = \{\tau_i^t \mid i \in C^t\}$.

3.2.2 Structure Advisor. When the flight stage changes, the Advisor retrieves the K most recent trajectory from the previous stages:

$$\mathcal{H}_K^{t+1} = \text{RecentK} \left(\{\tau_i^u \mid 1 \leq u \leq t, i \in C^u\} \right). \quad (14)$$

Trajectory-Guided Structure Reasoning. Design trajectories across continuous flight stages capture the temporal evolution of control parameters. The trajectories are combined with the knowledge of the dynamics of flight stage $\mathcal{K}_{\text{DYN}}$, the DLVAC description $\mathcal{D}_{\text{DLVAC}}$, and the task description $\mathcal{D}_{\text{task}}$ to construct the Advisor prompt $Q_A^{t+1} = \text{Prompt}(\mathcal{K}_{\text{DYN}}, \mathcal{D}_{\text{DLVAC}}, \mathcal{D}_{\text{task}}, \mathcal{H}_K^{t+1})$. The detailed prompt template is shown in Appendix D.

Our prompt guides the agent to understand the continuity of the flight process and the composition of trajectory. Then, the agent exploits historical trajectories to generate a promising initial structure search state for a new flight stage. The output of Advisor is decomposed into a rank matrix and a set of structure individuals:

$$(\mathbf{R}_1^{t+1}, \widehat{\mathbf{P}}_{S,1}^{t+1}) = \Pi_{\text{Advisor}}(Q_A^{t+1}), \quad (15)$$

where $\mathbf{R}_1^{t+1} \in \mathbb{R}^{4 \times N_{\text{exp}}}$ and $\widehat{\mathbf{P}}_{S,1}^{t+1} = [\widehat{\mathbf{n}}_{1,1}^{t+1}, \dots, \widehat{\mathbf{n}}_{M',1}^{t+1}]^T \in \mathbb{Z}_{>0}^{M' \times 4}$, M' denotes the number of individuals generated by the Advisor.

3.2.3 Search Reinitialization. The individuals generated by the Advisor are used to initialize the structure population. When $M' < M$, the remaining individuals are randomly generated by

$$\mathbf{P}_{S,1}^{t+1} = \begin{bmatrix} \widehat{\mathbf{P}}_{S,1}^{t+1} \\ \mathbf{P}_{S,\text{rand}}^{t+1} \end{bmatrix} \in \mathbb{Z}_{>0}^{M \times 4}, \quad (16)$$

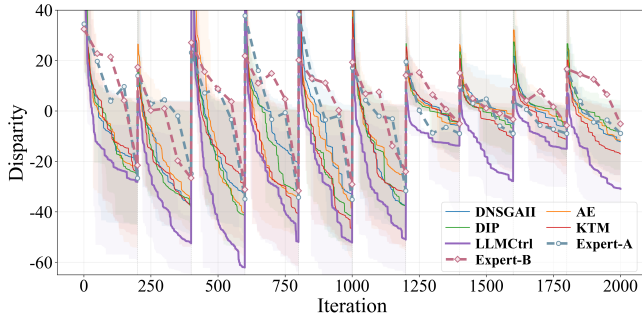
where $\mathbf{P}_{S,\text{rand}}^{t+1} \in \mathbb{Z}_{>0}^{(M-M') \times 4}$ used to restore the population size and preserve population diversity. For each $\mathbf{n}_{m,1}^{t+1}$, its activation matrix $\mathbf{A}_{m,1}^{t+1}$ is constructed from $\mathbf{R}_1^{t+1}$ following Equation (9). The variable-structure population in the new stage is then constructed as

$$\mathcal{P}_{X,1}^{t+1} = \left\{ \mathbf{x}_{m,1}^{t+1} = \text{Construct} \left(\mathbf{n}_{m,1}^{t+1}, \mathbf{P}_{V,1}^{t+1}; \mathbf{A}_{m,1}^{t+1} \right) \right\}_{m=1}^M. \quad (17)$$

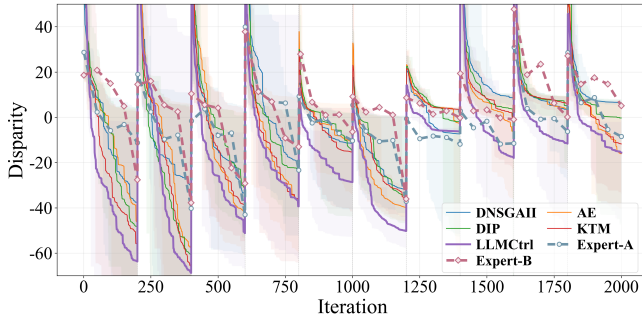
The reinitialized population provides promising structures and retains random individuals for exploration. The Structure Advisor is

Table 2: Overall performance on five DLVAC tasks under a change frequency of 200. Mean and standard deviation over ten runs are reported. Domain Experts tuning follows the runtime budget of LLMCtrl.

Tasks	Metrics	Optimization Algorithms				Domain Experts		LLMCtrl
		DNSGAI	AE	DIP	KTM	Expert-A	Expert-B	
Task-1	MSR $\uparrow$	64.28 $\pm$ 2.44	70.46 $\pm$ 3.80	65.34 $\pm$ 5.13	71.09 $\pm$ 3.45	71.05 $\pm$ 2.73	70.97 $\pm$ 2.92	81.73$\pm$3.33
	MD $\downarrow$	-7.05 $\pm$ 11.06	-10.81 $\pm$ 6.68	-7.50 $\pm$ 10.83	-8.71 $\pm$ 9.32	-8.87 $\pm$ 12.96	-7.53 $\pm$ 14.27	-13.53$\pm$8.46
	MHV $\uparrow$	9.44 $\pm$ 5.94	11.74 $\pm$ 3.87	10.16 $\pm$ 5.55	8.33 $\pm$ 3.15	9.98 $\pm$ 4.05	10.73 $\pm$ 5.89	17.48$\pm$2.02
Task-2	MSR $\uparrow$	56.63 $\pm$ 4.73	63.54 $\pm$ 3.63	58.29 $\pm$ 4.43	62.68 $\pm$ 4.49	71.80 $\pm$ 3.43	69.75 $\pm$ 4.95	81.54$\pm$1.84
	MD $\downarrow$	-5.49 $\pm$ 13.03	-9.22 $\pm$ 10.10	-7.75 $\pm$ 11.13	-6.32 $\pm$ 15.14	-6.94 $\pm$ 10.31	-5.59 $\pm$ 16.64	-11.43$\pm$14.38
	MHV $\uparrow$	9.45 $\pm$ 4.37	12.75 $\pm$ 4.80	9.64 $\pm$ 5.25	10.77 $\pm$ 5.33	12.75 $\pm$ 5.46	10.08 $\pm$ 4.19	13.67$\pm$4.90
Task-3	MSR $\uparrow$	51.44 $\pm$ 3.86	57.62 $\pm$ 4.60	53.46 $\pm$ 4.30	59.33 $\pm$ 3.95	69.30 $\pm$ 4.06	65.22 $\pm$ 4.47	77.26$\pm$2.74
	MD $\downarrow$	1.24 $\pm$ 18.45	-3.95 $\pm$ 15.97	-0.09 $\pm$ 16.03	-2.77 $\pm$ 15.61	-1.21 $\pm$ 17.12	-1.06 $\pm$ 15.41	-4.53$\pm$19.67
	MHV $\uparrow$	3.10 $\pm$ 3.43	3.97 $\pm$ 2.20	3.69 $\pm$ 7.19	4.12 $\pm$ 2.53	3.90 $\pm$ 1.69	3.67 $\pm$ 2.42	5.03$\pm$6.43
Task-4	MSR $\uparrow$	55.30 $\pm$ 5.08	63.52 $\pm$ 4.14	59.46 $\pm$ 3.57	62.60 $\pm$ 4.32	70.70 $\pm$ 3.91	68.83 $\pm$ 4.10	81.23$\pm$2.71
	MD $\downarrow$	-6.50 $\pm$ 12.26	-10.23 $\pm$ 13.44	-8.70 $\pm$ 10.68	-8.16 $\pm$ 14.63	-7.89 $\pm$ 14.48	-6.48 $\pm$ 16.52	-12.22$\pm$13.96
	MHV $\uparrow$	4.83 $\pm$ 2.08	6.09 $\pm$ 4.12	5.64 $\pm$ 3.10	5.83 $\pm$ 4.56	5.95 $\pm$ 2.53	5.35 $\pm$ 3.13	6.37$\pm$1.95
Task-5	MSR $\uparrow$	52.50 $\pm$ 4.41	55.46 $\pm$ 4.70	50.94 $\pm$ 3.28	59.98 $\pm$ 5.15	67.60 $\pm$ 3.45	62.40 $\pm$ 4.73	78.44$\pm$3.54
	MD $\downarrow$	0.30 $\pm$ 18.54	-5.23 $\pm$ 15.40	-2.03 $\pm$ 15.19	-3.97 $\pm$ 15.53	-1.94 $\pm$ 16.26	-2.32 $\pm$ 15.03	-5.25$\pm$19.27
	MHV $\uparrow$	1.86 $\pm$ 1.70	2.27 $\pm$ 1.19	2.05 $\pm$ 1.45	2.94 $\pm$ 1.38	2.87 $\pm$ 1.63	1.58 $\pm$ 1.52	3.52$\pm$5.78



(a) The disparity value comparison of Task-2



(b) The disparity value comparison of Task-3

Figure 4: Disparity curves under a change frequency of 200.

invoked only upon flight stage changes, whereas Structure Designer interactions remain controlled by the diversity criterion.

4 Experiments

We conduct extensive experiments with particular emphasis on comparing LLMCtrl with domain experts. Specifically, we investigate the following research questions (RQs): **RQ1:** Can LLMCtrl

outperform domain experts and optimization algorithms on DLVAC tasks? **RQ2:** What are the contributions of the proposed mechanisms, and which component is most critical? **RQ3:** How sensitive is LLMCtrl to different hyperparameter settings?

4.1 Experimental Settings

4.1.1 DLVAC Tasks. Table 1 summarizes five DLVAC tasks derived from real-world flight data and their corresponding target thresholds. For each task, we set $T = 10$ flight stages, with 30, 26, 27, 27, and 27 transfer functions in the initial stage. The change frequency is set to 200 and 800, denoting iterations between stage changes. The change severities are set to 2, 2, 3, 4, and 3, specifying the change in the number of transfer functions at each stage transition.

4.1.2 Evaluation Metrics. We evaluate the control performance using: (i) Mean Success Rate (MSR), which measures the proportion of transfer functions satisfying all targeted thresholds; (ii) Mean Disparity (MD), which quantifies the deviation of the obtained solutions from targeted thresholds; and (iii) Mean Hypervolume [18] (MHV), which evaluates the diversity of the obtained solution. A higher MSR indicates better control ability, a lower MD reflects closer control disparity, and a higher MHV represents superior Pareto dominance. Metric definitions are provided in **Appendix E.1**.

4.1.3 Competing Methods. We consider six competing methods to cover both the optimization algorithms and the domain expert design. We adapt (i) DNSGAI [7], (ii) AE-DMOEA [10] (AE), (iii) DIP-DMOEA [40] (DIP), and (iv) KTM-DMOEA [20] (KTM) to the DLVAC setting; (v) Expert-A is a senior engineer with nearly ten years of design experience, and (vi) Expert-B is a researcher with three years of project experience. Expert design relies on proprietary software and manual interpretation of Bode plots. Detailed adaptations for algorithms are provided in **Appendix E.2**.

4.1.4 Implement Details. The DLVAC tasks are implemented by domain experts using real-world flight data. LLMCtrl runs on our

Table 3: Overall performance on five DLVAC tasks under a change frequency of 800. Mean and standard deviation over ten runs are reported. Domain Experts tuning follows the runtime budget of LLMCtrl.

Tasks	Metrics	Optimization Algorithms				Domain Experts		LLMCtrl
		DNSGAI	AE	DIP	KTM	Expert-A	Expert-B	
Task-1	MSR $\uparrow$	69.33 $\pm$ 2.13	72.67 $\pm$ 2.72	75.71 $\pm$ 3.04	78.62 $\pm$ 1.66	78.85 $\pm$ 2.25	76.37 $\pm$ 3.92	92.41$\pm$1.67
	MD $\downarrow$	-19.37 $\pm$ 4.57	-22.38 $\pm$ 6.21	-18.64 $\pm$ 4.23	-23.57 $\pm$ 6.09	-21.59 $\pm$ 6.38	-19.72 $\pm$ 7.89	-23.88$\pm$7.66
	MHV $\uparrow$	10.42 $\pm$ 5.83	13.94 $\pm$ 3.64	11.86 $\pm$ 5.46	14.51 $\pm$ 4.17	12.21 $\pm$ 3.97	10.97 $\pm$ 5.16	20.90$\pm$9.33
Task-2	MSR $\uparrow$	70.11 $\pm$ 3.63	73.64 $\pm$ 3.01	72.15 $\pm$ 2.70	75.41 $\pm$ 2.68	76.70 $\pm$ 3.57	75.93 $\pm$ 1.84	91.22$\pm$2.04
	MD $\downarrow$	-20.54 $\pm$ 9.11	-24.51 $\pm$ 9.80	-25.17 $\pm$ 7.13	-24.41 $\pm$ 11.93	-21.92 $\pm$ 12.34	-27.04 $\pm$ 11.09	-31.27$\pm$10.61
	MHV $\uparrow$	12.64 $\pm$ 4.29	14.68 $\pm$ 4.50	15.87 $\pm$ 5.44	14.02 $\pm$ 4.71	14.53 $\pm$ 5.34	13.32 $\pm$ 3.48	17.42$\pm$4.64
Task-3	MSR $\uparrow$	68.47 $\pm$ 3.68	71.81 $\pm$ 4.73	67.81 $\pm$ 4.10	73.84 $\pm$ 3.74	74.30 $\pm$ 3.29	74.89 $\pm$ 4.58	87.90$\pm$2.03
	MD $\downarrow$	-29.53 $\pm$ 8.95	-34.92 $\pm$ 10.32	-28.23 $\pm$ 11.85	-35.22 $\pm$ 12.85	-31.40 $\pm$ 10.26	-30.16 $\pm$ 14.82	-38.22$\pm$13.62
	MHV $\uparrow$	6.87 $\pm$ 3.54	8.32 $\pm$ 2.60	7.38 $\pm$ 4.71	8.22 $\pm$ 4.38	9.01 $\pm$ 2.41	7.72 $\pm$ 4.12	10.39$\pm$4.90
Task-4	MSR $\uparrow$	70.82 $\pm$ 3.41	73.20 $\pm$ 2.01	69.93 $\pm$ 3.57	74.60 $\pm$ 2.13	74.95 $\pm$ 2.74	77.86 $\pm$ 5.08	92.88$\pm$1.41
	MD $\downarrow$	-23.05 $\pm$ 10.78	-27.53 $\pm$ 11.61	-26.98 $\pm$ 9.39	-28.97 $\pm$ 13.34	-26.01 $\pm$ 10.72	-28.93 $\pm$ 11.49	-33.91$\pm$9.39
	MHV $\uparrow$	6.20 $\pm$ 2.09	8.16 $\pm$ 1.53	7.29 $\pm$ 3.03	8.29 $\pm$ 3.48	9.57 $\pm$ 2.46	8.30 $\pm$ 2.10	11.26$\pm$3.27
Task-5	MSR $\uparrow$	70.51 $\pm$ 2.56	71.63 $\pm$ 4.31	73.27 $\pm$ 3.19	76.47 $\pm$ 5.62	72.41 $\pm$ 2.67	70.05 $\pm$ 2.06	88.60$\pm$3.52
	MD $\downarrow$	-32.81 $\pm$ 8.98	-35.59 $\pm$ 10.23	-29.35 $\pm$ 11.57	-34.38 $\pm$ 9.79	-32.79 $\pm$ 13.75	-31.77 $\pm$ 12.64	-36.55$\pm$14.04
	MHV $\uparrow$	6.83 $\pm$ 4.09	5.98 $\pm$ 2.70	7.38 $\pm$ 1.41	7.21 $\pm$ 3.15	8.16 $\pm$ 2.79	6.74 $\pm$ 3.15	8.63$\pm$4.32

Table 4: Comparison of variable-structure control parameters designed by LLMCtrl and Domain Experts.

Methods	Control Parameter Structure (N_1, N_2, N_3, N_4)				
	T1-S3	T2-S4	T3-S5	T4-S6	T5-S9
LLMCtrl	(2,6,1,7)	(2,7,2,8)	(3,9,1,11)	(4,12,3,13)	(3,8,2,9)
Expert-A	(1,5,1,5)	(1,5,2,5)	(2,4,2,6)	(2,7,2,7)	(2,5,1,6)
Expert-B	(2,4,2,4)	(1,6,2,8)	(2,6,2,7)	(1,8,1,9)	(1,6,1,7)

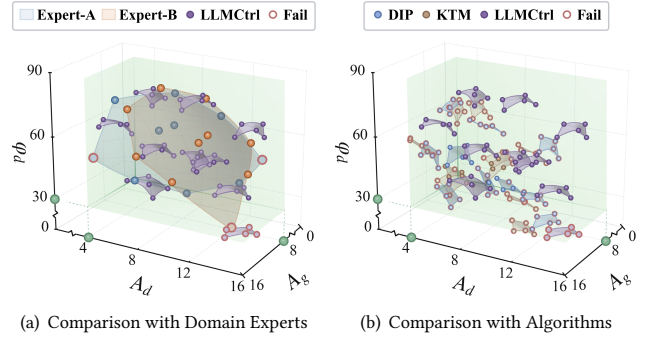
independently developed tool and accesses LLM agents through external APIs. All optimization algorithms use the same population size of $M=150$ to ensure fairness. We set $\epsilon_S = 0.3$ for invoking Structure Designer and retain $K=30$ design trajectories for Structure Advisor. The experiments are performed on a computer with an Intel(R) Core(TM) i7-10700 CPU and 64 GB of RAM.

4.2 RQ1: Overall Performance on DLVAC Tasks

We evaluate LLMCtrl with competing methods under different change frequencies, as shown in Tables 2 and 3. MSR is reported in (%), MHV is scaled by 10^3 , 10^6 , 10^8 , 10^5 , and 10^7 , respectively. The results show that LLMCtrl consistently outperforms domain experts and optimization algorithms across all DLVAC tasks.

4.2.1 Control Ability. LLMCtrl achieves the highest MSR under both change frequencies. Notably, relative to Expert-A, LLMCtrl improves the average MSR across the five tasks by 9.95% at a change frequency of 200 and by 15.16% at 800, delivering an average improvement of **12.56%**. This evidence suggests that LLMCtrl can outperform an experienced expert in control parameter design within the same design time in practice. **Appendix E.3** provides a detailed control performance comparison via Bode plots.

We further analyze the structural differences between the control parameters designed by LLMCtrl and those by domain experts, as summarized in Table 4. T(.) and S(.) denote tasks and flight stages,

**Figure 5: Control parameters of Task-1 on rigid component indicators under a change frequency of 800.**

respectively. We present the experts' selection of LLMCtrl solutions from the Pareto set whose structures are notably different from their experience under a change frequency of 800. LLMCtrl tends to construct more complex structures, demonstrating its ability to move beyond expert preferences and uncover alternative structures that may not be readily identified through expert experience alone.

4.2.2 Control Disparity. Figure 4 presents the disparity curves of Task-2 and Task-3 under a change frequency of 200. Given the same time budget, we report the results of five parameter designs produced by the experts at each flight stage. Consistent with the MD results in Table 2, the optimization algorithms without domain knowledge fail to outperform the expert designs. In contrast, with the assistance of LLM agents, LLMCtrl demonstrates a strong capability to search for better control parameters.

4.2.3 Pareto Quality. We select Task-1 to visually examine the Pareto dominance of the control parameters obtained by LLMCtrl. Low-Freq amp A_d , High-Freq amp A_g , and low-freq phase ϕ_d constitute the rigid component indicators in DLVAC. Figure 5 plots

Table 5: Ablation Study under a change frequency of 200. MSR is reported in (%), MHV is scaled by 10^3 , 10^6 , 10^8 , 10^5 , and 10^7 .

Variants	Task-1			Task-2			Task-3			Task-4			Task-5		
	MSR $\uparrow$	MD $\downarrow$	MHV $\uparrow$	MSR $\uparrow$	MD $\downarrow$	MHV $\uparrow$	MSR $\uparrow$	MD $\downarrow$	MHV $\uparrow$	MSR $\uparrow$	MD $\downarrow$	MHV $\uparrow$	MSR $\uparrow$	MD $\downarrow$	MHV $\uparrow$
w/o Designer	71.07	-8.60	9.26	69.33	-8.03	9.01	70.63	-1.44	2.87	67.21	-7.51	2.31	64.89	-2.59	1.25
LLMCtrl-RR	75.31	-11.42	13.64	73.53	-10.31	13.22	72.47	-3.13	1.05	71.34	-7.61	4.09	68.25	-4.21	1.79
LLMCtrl-PI	75.92	-12.94	14.81	74.05	-11.05	12.96	73.97	-1.81	3.23	71.82	-8.98	3.85	67.73	-4.53	2.91
w/o Advisor	72.19	-9.68	10.15	70.94	-8.87	10.36	70.11	-2.08	1.54	68.76	-6.95	1.79	67.42	-3.63	2.87
LLMCtrl-PT	77.53	-12.79	13.68	73.42	-11.26	12.41	73.31	-3.78	2.21	72.18	-9.11	3.29	69.11	-5.08	2.36
LLMCtrl-AT	77.86	-10.53	13.95	74.75	-10.87	12.64	73.64	-3.51	3.47	71.51	-9.34	4.52	70.42	-4.84	2.63
LLMCtrl	81.73	-13.53	17.48	81.54	-11.43	13.67	77.26	-4.53	5.03	81.23	-12.22	6.37	78.44	-5.25	3.52

the control parameters obtained by the competing methods across ten flight stages; fail represents an unsuccessful control. For each stage, we select five solutions from each algorithm and one from each expert. Consistent with the MHV results in Table 3, LLMCtrl generates control parameters that dominate those of the competing methods across all three indicators at every flight stage.

4.3 RQ2: Ablation Study

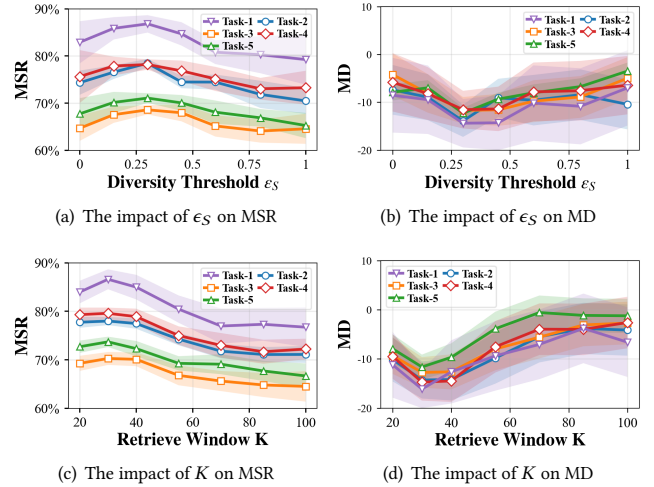
4.3.1 Ablation on Mechanisms. We construct several variants of LLMCtrl to investigate the contributions of the designed mechanism: (i) w/o Designer adopts the same population form as the competing algorithms and retains only Pareto solutions in the design trajectories; (ii) LLMCtrl-RR replaces the generated rank matrix with a random one; (iii) LLMCtrl-PI invokes the Designer periodically rather than a diversity trigger; (iv) w/o Advisor randomly reinitializes the population after each stage change; (v) LLMCtrl-PT retains only Pareto individuals in the design trajectories; and (vi) LLMCtrl-AT retains only activation matrices.

As reported in Table 5, the results reveal that both proposed mechanisms are indispensable. For the LLM-Assisted Coevolution Activation Mechanism, a random rank matrix cannot reproduce the structural guidance provided by the Designer. Periodic invocation improves performance to some extent, but frequent intervention can interrupt the evolution process and introduce unnecessary changes. By contrast, diversity-triggered invocation provides guidance only when the population loses sufficient structural variation, making it a more principled intervention strategy. For the LLM-Assisted Search Reinitialization Mechanism, removing cross-stage knowledge leads to a substantial loss in control performance. The degradation observed when retaining only Pareto individuals or only activation matrices suggests that these two components preserve different latent historical design experience and provide complementary guidance for subsequent flight stages.

4.3.2 Ablation on LLM APIs. Table 6 reports the performance of LLMCtrl under a change frequency of 800 with five LLM backbones: (i) Llama4 (Llama-4-Maverick-17B-128E-Instruct), (ii) Qwen3 (Qwen3-235B-A22B-Instruct-2507), (iii) Deepseek4 (DeepSeek-V4-Pro), (iv) GPT4 (GPT-4.1-2025-04-14), and (v) GPT5 (GPT-5.5-2026-04-23). It is worth noting that different LLM APIs may affect LLMCtrl’s performance to some extent, but they do not cause unstable or severely degraded behavior. The key reason is that agents are not responsible for directly optimizing the control parameters, they only provide high-level guidance for structure design and search

Table 6: Results of LLMCtrl with different LLM APIs.

Tasks	Metrics	Llama	Qwen3	DeepSeek	GPT-4	GPT-5
Task-1	MSR $\uparrow$	88.56	79.92	86.13	91.04	92.41
	MD $\downarrow$	-27.42	-21.18	-25.55	-24.37	-23.88
Task-2	MSR $\uparrow$	89.88	83.05	85.67	88.31	91.22
	MD $\downarrow$	-26.08	-25.23	-25.41	-24.19	-31.27
Task-3	MSR $\uparrow$	85.02	78.76	79.89	88.55	87.90
	MD $\downarrow$	-32.92	-28.72	-25.45	-35.64	-38.22
Task-4	MSR $\uparrow$	94.34	77.91	80.21	90.68	92.88
	MD $\downarrow$	-32.27	-25.85	-24.93	-29.86	-33.91
Task-5	MSR $\uparrow$	84.15	80.83	73.56	86.72	88.60
	MD $\downarrow$	-28.74	-21.31	-23.46	-27.52	-36.55

**Figure 6: Sensitivity of LLMCtrl to the diversity threshold ϵ_S and retrieve window K on DLVAC tasks.**

reinitialization. Therefore, the optimization process can reduce LLMCtrl’s sensitivity to variations in LLM outputs.

4.4 RQ3: Hyperparameter Sensitivity

Our sensitivity analysis examines two key hyperparameters in LLMCtrl: (i) the diversity threshold ϵ_S for Designer invocation and (ii) the retrieve window K for Advisor historical design trajectories. Figure 6 reports the sensitivity of MSR and MD to the two

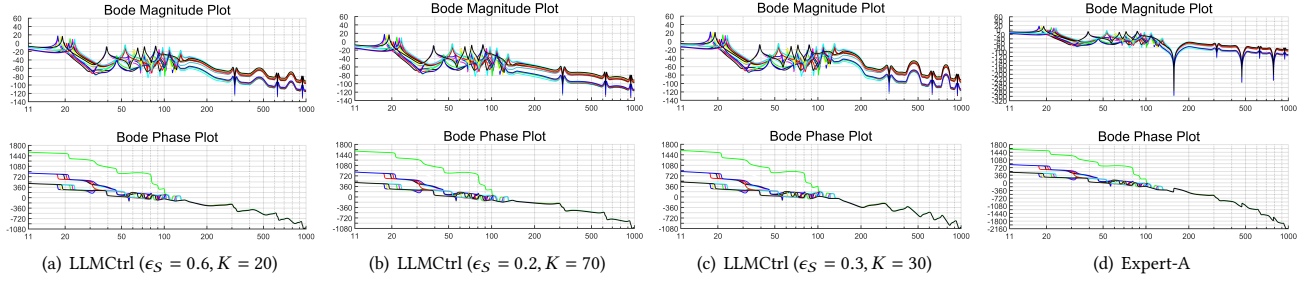


Figure 7: Bode plots comparing different LLMCtrl configurations with Expert-A on Task-4 at Stage-7 under a change frequency of 800. The axes report frequency in (rad/s), magnitude in (dB), and phase in ($^{\circ}$).

hyperparameters; Figure 7 visualizes the control performance of different hyperparameter configurations via Bode plots. We draw two observations: (1) evolutionary populations tend to lose structural diversity as the search concentrates on Pareto-dominant regions. By setting $\epsilon_S = 0.3$, LLMCtrl permits moderate population homogenization while triggering the Designer when diversity loss occurs. The resulting intervention introduces alternative structures and helps the population escape the current dominance pattern. (2) the Designer is invoked in approximately 2.7% of all iterations across the flight process. The settings of K should be coordinated with ϵ_S . Frequent Designer invocations under a large ϵ_S require a larger K for cross-stage trajectories; a small ϵ_S may make the retrieval window encompass all past trajectories. Although $K = 30$ may initially cover past trajectories, it retains adequate historical guidance while avoiding excessive intervention in the ongoing value search.

5 Related Works

5.1 Dynamic Launch Vehicle Attitude Control

The growing complexity of launch vehicle configurations and task environments demands attitude control systems to remain stable and robust under substantial variations in system dynamics [23, 42]. At the control level, existing research has explored incremental non-linear control allocation, adaptive neural control, and reinforcement learning for complex flight vehicle models [3, 4, 6, 14, 25, 35, 43]. At the dynamic level, optimization algorithms [38, 39] have been developed to cope with changing environments via autoencoding-based population prediction [10], directional improvement learning [40], historical knowledge selection [15], and knowledge transfer and maintenance [20]. However, existing methods typically assume decision spaces of fixed dimensionality and therefore struggle to handle structural variations that reshape the search space.

5.2 LLMs for Dynamic Control Adaptation

In recent years, LLMs have demonstrated strong capabilities in contextual understanding, domain knowledge integration, and multi-step reasoning [21, 29]. The intersection of LLMs and optimization has rapidly emerged and attracted increasing attention [12, 24, 34, 37]. Building on these advances, LLMs have recently been introduced into control design and optimization [5, 13, 30, 47]. Several studies further employ LLMs for controller adaptation and adaptive compensator design under uncertain system [33, 44]. These

developments have further motivated the application of LLMs to aerospace engineering [16, 49]. However, the structural variation of control parameters in DLVAC makes it difficult for existing methods to identify transferable design knowledge across stages and develop effective mechanisms for exploiting such knowledge.

6 Limitations and Ethical Considerations

Limitations. Our limitations include: (i) agent invocation increases runtime relative to optimization algorithms; (ii) prompts cannot fully replicate the deep expertise of domain experts; and (iii) data sensitivity prevents domain-specific fine-tuning, leaving LLMCtrl solely reliant on CoT reasoning to assist control parameter design.

Ethical Considerations. LLMCtrl is proposed exclusively as an engineering assistance tool for DLVAC, and its outputs must undergo safety validation before practical use. Potential dual-use risks further require appropriate oversight and safeguards.

7 Generative AI Usage

We used generative AI models in two ways: (i) as functional components of the LLMCtrl mechanisms, as described in the methodology section; and (ii) solely to improve linguistic clarity and readability during manuscript preparation. All technical ideas, methodological designs, experiments, analyses, and conclusions were developed, validated, and reviewed by the authors.

8 Conclusion

This study proposes LLMCtrl, a novel framework that employs LLM agents to assist control parameter design in DLVAC. LLMCtrl establishes a new paradigm for control parameter design through two complementary mechanisms: LLM-Assisted Coevolution Activation and LLM-Assisted Search Reinitialization. These mechanisms emulate expert design procedures by exploring diverse control parameter structures within flight stages and identifying promising structures for reuse across flight stages. Extensive experiments on five DLVAC tasks reveal the answer: LLMCtrl can outperform domain experts within an equal control parameter design time budget. Furthermore, LLMCtrl demonstrates superior performance over state-of-the-art optimization algorithms. Our future work aims to investigate robust control parameter design that remains effective under uncertain equipment failures during launch vehicle flight.

References

- [1] Tasnim Ahmed and Salimur Choudhury. 2024. LM4OPT: Unveiling the potential of Large Language Models in formulating mathematical optimization problems. *INFOR: Information Systems and Operational Research* 62, 4 (2024), 559–572.
- [2] Boris Benedikter, Alessandro Zavoli, Guido Colasurdo, Simone Pizzurro, and Enrico Cavallini. 2022. Convex optimization of launch vehicle ascent trajectory with heat-flux and splash-down constraints. *Journal of Spacecraft and Rockets* 59, 3 (2022), 900–915.
- [3] Chengyu Cao, Fanbiao Li, Rong Ding, Tingwen Huang, Chunhua Yang, and Weihua Gui. 2025. Intelligent Attitude Control for Morphing Flight Vehicle: A Deep Reinforcement Learning Approach. *IEEE Transactions on Vehicular Technology* (2025).
- [4] Shanelle G Clarke and Inseok Hwang. 2020. Deep reinforcement learning control for aerobatic maneuvering of agile fixed-wing aircraft. In *AIAA Scitech 2020 Forum*. 0136.
- [5] Chenggang Cui, Jiaming Liu, Peifeng Hui, Pengfeng Lin, and Chuanlin Zhang. 2025. GenControl: Generative AI-Driven Autonomous Design of Control Algorithms. *arXiv preprint arXiv:2506.12554* (2025).
- [6] Pepijn de Heer, Coen C de Visser, Marijn L Hoogendoorn, and Henk W Jentink. 2023. Incremental nonlinear control allocation for an aircraft with distributed electric propulsion. In *AIAA SCITECH 2023 Forum*. 1248.
- [7] Kalyanmoy Deb, Udaya Bhaskara Rao N, and Sindhya Karthik. 2007. Dynamic multi-objective optimization and decision-making using modified NSGA-II: A case study on hydro-thermal power scheduling. In *International conference on evolutionary multi-criterion optimization*. Springer, 803–817.
- [8] Xiaoshuai Fan, Xibin Bai, Zhenyu Jiang, and Shifeng Zhang. 2024. An optimization method of attitude control parameters based on genetic algorithm for the boost-glide rocket. *IEEE Trans. Aerospace Electron. Systems* 60, 4 (2024), 4356–4376.
- [9] Guhao Feng, Bohang Zhang, Yuntian Gu, Haotian Ye, Di He, and Liwei Wang. 2023. Towards revealing the mystery behind chain of thought: a theoretical perspective. *Advances in Neural Information Processing Systems* 36 (2023), 70757–70798.
- [10] Liang Feng, Wei Zhou, Weichen Liu, Yew-Soon Ong, and Kay Chen Tan. 2020. Solving dynamic multiobjective problem via autoencoding evolutionary search. *IEEE Transactions on Cybernetics* 52, 5 (2020), 2649–2662.
- [11] Yuan GAO, Yongliang TIAN, Hu LIU, and Xue SUN. 2020. Gaussian fitting based optimal design of aircraft mission success space using multi-objective genetic algorithm. *Chinese Journal of Aeronautics* 33, 12 (2020), 3318–3330. doi:10.1016/j.cja.2020.05.013
- [12] Qingyan Guo, Rui Wang, Junliang Guo, Bei Li, Kaitao Song, Xu Tan, Guoqing Liu, Jiang Bian, and Yujiu Yang. 2024. Connecting Large Language Models with Evolutionary Algorithms Yields Powerful Prompt Optimizers. In *The Twelfth International Conference on Learning Representations*. <https://openreview.net/forum?id=ZG3RaNIso8>
- [13] Xingang Guo, Darioush Keivan, Usman Syed, Lianhui Qin, Huan Zhang, Geir Dullerud, Peter Seiler, and Bin Hu. 2024. Controlagent: Automating control system design via novel integration of llm agents and domain expertise. *arXiv preprint arXiv:2410.19811* (2024).
- [14] Jiong He, Binwu Ren, Yousong Xu, Qijun Zhao, Siliang Du, and Bo Wang. 2025. Neural network adaptive attitude control of full-states quad tiltrotor UAV. *Aerospace* 12, 8 (2025), 684.
- [15] Yaru Hu, Jinhua Zheng, Shouyong Jiang, Shengxiang Yang, Juan Zou, and Rui Wang. 2023. A mahalanobis distance-based approach for dynamic multiobjective optimization with stochastic changes. *IEEE Transactions on Evolutionary Computation* 28, 1 (2023), 238–251.
- [16] Amit Jain and Richard Linares. 2026. Autonomous Reasoning for Spacecraft Control: A Large Language Model Framework with Group Relative Policy Optimization. In *2026 IEEE Aerospace Conference*. 1–14.
- [17] Caigao Jiang, Xiang Shu, Hong Qian, Xingyu Lu, Jun Zhou, Aimin Zhou, and Yang Yu. 2024. LLMOPT: Learning to Define and Solve General Optimization Problems from Scratch. *arXiv preprint arXiv:2410.13213* (2024).
- [18] Shouyong Jiang, Marcus Kaiser, Shengxiang Yang, Stefanos Kollias, and Natalio Krasnogor. 2019. A scalable test suite for continuous dynamic multiobjective optimization. *IEEE transactions on cybernetics* 50, 6 (2019), 2814–2826.
- [19] Shouyong Jiang, Juan Zou, Shengxiang Yang, and Xin Yao. 2022. Evolutionary dynamic multi-objective optimisation: A survey. *Comput. Surveys* 55, 4 (2022), 1–47.
- [20] Qiuzhen Lin, Yulong Ye, Lijia Ma, Min Jiang, and Kay Chen Tan. 2023. Dynamic multiobjective evolutionary optimization via knowledge transfer and maintenance. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 54, 2 (2023), 936–949.
- [21] Chen Ling, Xujiang Zhao, Jiaying Lu, Chengyuan Deng, Can Zheng, Junxiang Wang, Tanmoy Chowdhury, Yun Li, Hejie Cui, Xuchao Zhang, et al. 2025. Domain specialization as the key to make large language models disruptive: A comprehensive survey. *Comput. Surveys* 58, 3 (2025), 1–39.
- [22] Fei Liu, Xialiang Tong, Mingxuan Yuan, Xi Lin, Fu Luo, Zhenkun Wang, Zhichao Lu, and Qingfu Zhang. 2024. Evolution of heuristics: Towards efficient automatic algorithm design using large language model, 2024. URL <https://arxiv.org/abs/2401.02051> (2024).
- [23] Hangxian Liu, Ningning Wang, Zhesuo Zhang, and Hui Yin. 2023. Agile and precise attitude control of tiltrotor aircraft in transition flight. *IEEE Transactions on Intelligent Vehicles* 9, 1 (2023), 787–798.
- [24] Shengcai Liu, Caishun Chen, Xinghua Qu, Ke Tang, and Yew-Soon Ong. 2024. Large Language Models as Evolutionary Optimizers. In *2024 IEEE Congress on Evolutionary Computation (CEC)*. 1–8. doi:10.1109/CEC60901.2024.10611913
- [25] Shiqian Liu, Weizhi Lyu, Qian Zhang, Congjie Yang, and James F Whidborne. 2025. Neural-network-based incremental backstepping sliding mode control for flying-wing aircraft. *Journal of Guidance, Control, and Dynamics* 48, 3 (2025), 600–614.
- [26] Shengzhao Pang, Heng Liu, Yingxue Chen, Bo Cheng, Zhaoyong Mao, and Yigeng Huangfu. 2025. Distributed Electric Propulsion Aircraft Attitude Control Based on Assisted Training Multi-Agent Reinforcement Learning. *IEEE Transactions on Transportation Electrification* (2025).
- [27] Gan Ruan, Zhanglu Hou, and Xin Yao. 2025. Coping With a Severely Changing Number of Objectives in Dynamic Multi-Objective Optimization. *IEEE Transactions on Evolutionary Computation* (2025).
- [28] Gan Ruan, Leandro L Minku, Stefan Menzel, Bernhard Sendhoff, and Xin Yao. 2024. Learning to expand/contract pareto sets in dynamic multi-objective optimization with a changing number of objectives. *IEEE Transactions on Evolutionary Computation* (2024).
- [29] Jiankai Sun, Chuanyang Zheng, Enze Xie, Zhengying Liu, Ruihang Chu, Jianing Qiu, Jiaqi Xu, Mingyu Ding, Hongyang Li, Mengzhe Geng, et al. 2025. A survey of reasoning with foundation models: Concepts, methodologies, and outlook. *Comput. Surveys* 57, 11 (2025), 1–43.
- [30] Kadir Tohma, Halil Ibrahim Okur, Handan Gürsoy-Demir, Merve Nilay Aydın, and Celaleddin Yeroğlu. 2025. SmartControl: Interactive PID controller design powered by LLM agents and control system expertise. *SoftwareX* 31 (2025), 102194.
- [31] Yao TONG, Mingqiang LUO, Shangqing REN, Zheng ZHANG, Chenguang XING, and Ziliang DU. 2025. A rapidly structured aircraft concept design method based on generative artificial intelligence. *Chinese Journal of Aeronautics* 38, 10 (2025). doi:10.1016/j.cja.2025.103629
- [32] Niki van Stein and Thomas Bäck. 2024. Llama: A large language model evolutionary algorithm for automatically generating metaheuristics. *IEEE Transactions on Evolutionary Computation* (2024).
- [33] Javal Vyas and Mehmet Mercangöz. 2025. Autonomous industrial control using an agentic framework with large language models. *IFAC-PapersOnLine* 59, 6 (2025), 349–354.
- [34] Xingyu Wu, Yan Zhong, Jibin Wu, Bingbing Jiang, and Kay Chen Tan. 2024. Large language model-enhanced algorithm selection: towards comprehensive algorithm representation. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (Jeju, Korea) (IJCAI '24)*. Article 579, 10 pages. doi:10.24963/ijcai.2024/579
- [35] Shihao Xu, Changzhu Wei, Ligen Cai, and Xiaorui Li. 2024. Neural network-based adaptive optimal tracking control for hypersonic morphing aircraft with appointed-time prescribed performance. *Journal of the Franklin Institute* 361, 12 (2024), 107026.
- [36] Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V Le, Denny Zhou, and Xinyun Chen. 2023. Large language models as optimizers. In *The Twelfth International Conference on Learning Representations*.
- [37] Shunyu Yao, Fei Liu, Xi Lin, Zhichao Lu, Zhenkun Wang, and Qingfu Zhang. 2025. Multi-objective evolution of heuristic using large language model. In *Proceedings of the Thirty-Ninth AAAI Conference on Artificial Intelligence and Thirty-Seventh Conference on Innovative Applications of Artificial Intelligence and Fifteenth Symposium on Educational Advances in Artificial Intelligence (AAAI'25/IAAI'25/EAAI'25)*. AAAI Press, Article 3024, 9 pages. doi:10.1609/aaai.v39i25.34922
- [38] Danial Yazdani, Ran Cheng, Donya Yazdani, Jürgen Branke, Yaochu Jin, and Xin Yao. 2021. A survey of evolutionary continuous dynamic optimization over two decades—Part A. *IEEE Transactions on Evolutionary Computation* 25, 4 (2021), 609–629.
- [39] Danial Yazdani, Ran Cheng, Donya Yazdani, Jürgen Branke, Yaochu Jin, and Xin Yao. 2021. A survey of evolutionary continuous dynamic optimization over two decades—Part B. *IEEE Transactions on Evolutionary Computation* 25, 4 (2021), 630–650.
- [40] Yulong Ye, Songbai Liu, Junwei Zhou, Qiuzhen Lin, Min Jiang, and Kay Chen Tan. 2024. Learning-based directional improvement prediction for dynamic multiobjective optimization. *IEEE Transactions on Evolutionary Computation* (2024).
- [41] Zhilong Yu, Yinghui Li, Maolong Lv, Binbin Pei, Wenfeng Xu, and Xiangwei Bu. 2024. Fixed-time adaptive fuzzy fault-tolerant attitude control for tailless aircraft without angular velocity measurements. *IEEE Transactions on Fuzzy Systems* 32, 4 (2024), 2458–2470.

- [42] Zhilong Yu, Yinghui Li, Maolong Lv, Binbin Pei, Wenfeng Xu, and Xiangwei Bu. 2024. Fixed-time adaptive fuzzy fault-tolerant attitude control for tailless aircraft without angular velocity measurements. *IEEE Transactions on Fuzzy Systems* 32, 4 (2024), 2458–2470.
- [43] Yuqi Yuan and Di Zhou. 2024. Reinforcement Learning for Dual-Control Aircraft Six-Degree-of-Freedom Attitude Control with System Uncertainty. *Aerospace* 11, 4 (2024), 281.
- [44] Rasoul Zahedifar, Mahdih Soleymani Baghshah, and Alireza Taheri. 2025. LLM-controller: Dynamic robot control adaptation using large language models. *Robotics and Autonomous Systems* 186 (2025), 104913.
- [45] Zhuosheng Zhang, Yao Yao, Aston Zhang, Xiangru Tang, Xinbei Ma, Zhiwei He, Yiming Wang, Mark Gerstein, Rui Wang, Gongshen Liu, et al. 2025. Igniting language intelligence: The hitchhiker's guide from chain-of-thought reasoning to language agents. *Comput. Surveys* 57, 8 (2025), 1–39.
- [46] Hongyu Zhou, Xiaogang Wang, Xun Li, and Naigang Cui. 2025. Trajectory planning for reusable launch vehicle with multisolution-based maximum entropy reinforcement learning. *IEEE Trans. Aerospace Electron. Systems* 61, 5 (2025), 11613–11627.
- [47] Zhongchao Zhou, Yuxi Lu, Yaonan Zhu, Yifan Zhao, Bin He, Liang He, Wenwen Yu, and Yusuke Iwasawa. 2025. LLMs-guided adaptive compensator: Bringing Adaptivity to Automatic Control Systems with Large Language Models. *arXiv preprint arXiv:2507.20509* (2025).
- [48] Zhihao Zhu, Yu Guo, and Zhi Gao. 2025. Optimization control of spacecraft proximation based on r-domination adaptive bare-bones particle swarm optimization algorithm. *Expert Systems with Applications* 278 (2025), 127269.
- [49] Enrico M Zucchelli, Di Wu, Julia Briden, Christian Hofmann, Victor Rodriguez-Fernandez, and Richard Linares. 2025. Fine-tuned language models as space systems controllers. *arXiv preprint arXiv:2501.16588* (2025).

A Flight Attitude Indicators

A.1 Variable-Structure Control Parameter

The control parameter considered in this work corresponds to a component of the compensator network used in launch vehicle control. For a linear time-invariant system, the compensator network is defined as the ratio of the Laplace transform of its output to that of its input. To facilitate understanding across disciplines and emphasize the central design problem in DLVAC, we use the more accessible term **control parameter** throughout this paper.

A.2 Bode plots

The frequency characteristics of $G_k^t(s)$ can be expressed as

$$\begin{cases} A_F(\omega) = 20\lg K + \sum_{i=1}^{N_1} 20\lg \sqrt{T_i^2 \omega^2 + 1} + \sum_{j=1}^{N_2} 20\lg \sqrt{f_d^A(\omega)} \\ \quad - \sum_{k=1}^{N_3} 20\lg \sqrt{T_k^2 \omega^2 + 1} - \sum_{z=1}^{N_4} 20\lg \sqrt{f_o^A(\omega)} \\ \varphi_F(\omega) = \sum_{i=1}^{N_1} \tan^{-1}(T_i \omega) + \sum_{j=1}^{N_2} \tan^{-1} f_d^\varphi(\omega) \\ \quad - \sum_{k=1}^{N_3} \tan^{-1}(T_k \omega) - \sum_{z=1}^{N_4} \tan^{-1} f_o^\varphi(\omega), \end{cases} \quad (18)$$

where $f_d^A(\omega)$ and $f_o^A(\omega)$ denote the second-order terms parameterized by (ω_n, ξ) , and $f_d^\varphi(\omega)$ and $f_o^\varphi(\omega)$ are the corresponding phase ratio terms. Here, $A_F(\omega)$ and $\varphi_F(\omega)$ represent the logarithmic amplitude- and phase-frequency characteristic of $G_k^t(s)$, respectively. Typically, we plot the curves of these two characteristics to obtain the logarithmic frequency response. Typically, we plot the curves of $A_F(\omega)$ and $\varphi_F(\omega)$ to obtain the logarithmic frequency response of the DLVAC system.

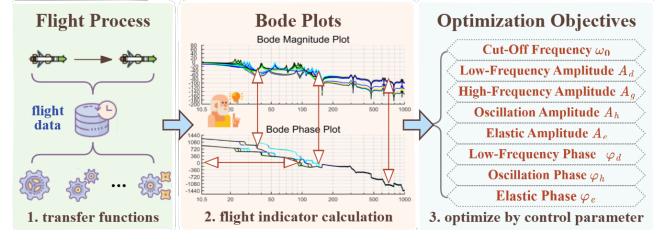


Figure 8: Illustration of the generation of launch vehicle flight attitude indicators.

The flight attitude indicators used to evaluate control parameters are extracted from the Bode plots generated by the corresponding transfer functions, following domain-specific rules. As shown in Figure 8, these indicators serve as the optimization objectives of the DLVAC tasks.

A.3 Domain-specific Rules

Based on the Bode plots, the flight control indicators of the control parameters are only determined using the following rules:

- **Cut-Off Frequency (ω_0):** The frequency point where the magnitude characteristic curve first crosses 0 dB downward within the entire frequency range.
- **Low-Frequency Amplitude (A_d):** The frequency point where the phase characteristic curve first crosses 180° upwards corresponds to the magnitude point on the magnitude-frequency characteristic curve.
- **High-Frequency Amplitude (A_g):** The frequency point where the phase characteristic curve first crosses 180° downwards corresponds to the magnitude point on the magnitude-frequency characteristic curve.
- **Oscillation Amplitude (A_h):** The magnitude point corresponding to the peak value on the magnitude characteristic curve.
- **Elastic Amplitude (A_e):** The magnitude point corresponding to the peak value on the magnitude characteristic curve.
- **Low-frequency Phase (φ_d):** The frequency point where the magnitude-frequency characteristic curve first crosses 0 dB corresponds to the phase point on the phase-frequency characteristic curve, with the difference from 180°.
- **Oscillation Phase (φ_h):** The frequency point where the magnitude-frequency characteristic curve crosses 0 dB corresponds to the phase point on the phase-frequency characteristic curve, with the absolute difference from 180°.
- **Elastic Phase (φ_e):** The frequency point where the magnitude-frequency characteristic curve crosses 0 dB corresponds to the phase point on the phase-frequency characteristic curve, with the absolute difference from 180°.

It is important to note that when multiple results are available, the maximum or minimum value is selected. While the rule for elastic gain is identical to that of oscillation gain, the frequency range and target values differ. Similarly, although the rule for the elastic phase margin is the same as for the oscillation phase, the specified frequency range and target values also vary.

Algorithm 1: LLMCtrl for Control Parameter Design

Input: Flight stages $\{\mathcal{E}^t\}_{t=1}^T$, population size M , iteration budgets $\{I_t\}_{t=1}^T$, diversity threshold ϵ_S , and trajectory window K

Output: Pareto control parameter sets $\{\mathcal{P}_{\text{Pareto}}^t\}_{t=1}^T$

```

1 for  $t \leftarrow 1$  to  $T$  do
2   Initial Structure Designer interaction index  $C^t \leftarrow \emptyset$ ;
3   Initial design trajectories repository  $\mathcal{T}^t \leftarrow \emptyset$ ;
4   if  $t = 1$  then
5      $(\mathbf{P}_{S,1}^t, \mathbf{P}_{V,1}^t) \leftarrow \text{InitializePopulations}(M)$ ;
6     /* LLM-Assisted Coevolution Activation */;
7      $Q_{D,1}^t \leftarrow \text{Prompt}(\mathcal{K}_V, \mathcal{D}_{\text{DLVAC}}, \mathcal{D}_{\text{task}}, \mathbf{P}_{S,1}^t, \mathbf{P}_{V,1}^t)$ ;
8     Generate the rank matrix  $\mathbf{R}_1^t \leftarrow \Pi_{\text{Designer}}(Q_{D,1}^t)$ ;
9      $C^t \leftarrow \{1\}$ ;
10  else
11    /* LLM-Assisted Search Reinitialization */;
12     $(\mathbf{R}_1^t, \mathbf{P}_{S,1}^t) \leftarrow \text{SearchReinitialize}(\{\mathcal{T}^u\}_{u=1}^{t-1}, \mathcal{D}_{\text{task}}, M, K)$ ;
13     $\mathbf{P}_{V,1}^t \leftarrow \text{ReinitializeValuePopulation}(M)$ ;
14  for  $i \leftarrow 1$  to  $I_t$  do
15     $\{\mathbf{A}_{m,i}^t\}_{m=1}^M \leftarrow \text{Activate}(\mathbf{R}_i^t, \mathbf{P}_{S,i}^t)$ ;
16     $\mathcal{P}_{X,i}^t \leftarrow \text{Construct}(\mathbf{P}_{S,i}^t, \mathbf{P}_{V,i}^t; \{\mathbf{A}_{m,i}^t\}_{m=1}^M)$ ;
17     $\mathcal{F}_i^t \leftarrow \text{Evaluate}(\mathcal{P}_{X,i}^t, \mathcal{E}^t)$  via Code plots;
18    if  $i \in C^t$  then
19      /* Design Trajectories Construction */;
20       $\mathcal{I}_{\mathcal{P},i}^t \leftarrow \{m \mid \mathbf{x}_{m,i}^t \in \text{Nondominated}(\mathcal{P}_{X,i}^t)\}$ ;
21       $\tau_i^t \leftarrow \{(\mathbf{A}_{m,i}^t, \mathbf{n}_{m,i}^t) \mid m \in \mathcal{I}_{\mathcal{P},i}^t\}$ ;
22       $\mathcal{T}^t \leftarrow \mathcal{T}^t \cup \{\tau_i^t\}$ ;
23    if  $i < I_t$  then
24      /* Multiobjective Coevolution */;
25       $(\mathbf{P}_{S,i+1}^t, \mathbf{P}_{V,i+1}^t) \leftarrow \text{Coevolve}(\mathbf{P}_{S,i}^t, \mathbf{P}_{V,i}^t, \mathcal{F}_i^t)$ ;
26       $\mathbf{R}_{i+1}^t \leftarrow \mathbf{R}_i^t$ ;
27       $d_{S,i+1}^t \leftarrow \text{Diversity}(\mathbf{P}_{S,i+1}^t)$ ;
28      if  $d_{S,i+1}^t < \epsilon_S$  then
29        /* LLM-Assisted Coevolution Activation */;
30         $Q_{D,i+1}^t \leftarrow \text{Prompt}(\mathcal{K}_V, \mathcal{D}_{\text{DLVAC}}, \mathcal{D}_{\text{task}}, \mathbf{P}_{S,i+1}^t, \mathbf{P}_{V,i+1}^t)$ ;
31         $\mathbf{R}_{i+1}^t \leftarrow \Pi_{\text{Designer}}(Q_{D,i+1}^t)$ ;
32         $C^t \leftarrow C^t \cup \{i+1\}$ ;
33   $\mathcal{P}_{\text{Pareto}}^t \leftarrow \text{Nondominated}(\mathcal{P}_{X,I_t}^t)$ 

```

B Pseudocode for our proposed framework

The pseudocode of our proposed LLMCtrl is described in Algorithm 1 and the pseudocode of LLM-Assisted Search Reinitialization is described in Algorithm 2.

Algorithm 2: LLM-Assisted Search Reinitialization

Input: Historical design trajectory repositories $\{\mathcal{T}^u\}_{u=1}^{t-1}$, task description $\mathcal{D}_{\text{task}}$, population size M , and trajectory window K

Output: Rank matrix $\mathbf{R}_1^t$ and structure population $\mathbf{P}_{S,1}^t$ for the new flight stage

```

1 Select design trajectories
    $\mathcal{H}_K^t \leftarrow \text{RecentK}(\{\tau_i^u \mid 1 \leq u < t, i \in C^u\}, K)$ ;
2  $Q_A^t \leftarrow \text{Prompt}(\mathcal{K}_{\text{DYN}}, \mathcal{D}_{\text{DLVAC}}, \mathcal{D}_{\text{task}}, \mathcal{H}_K^t)$ ;
3  $(\mathbf{R}_1^t, \hat{\mathbf{P}}_{S,1}^t) \leftarrow \Pi_{\text{Advisor}}(Q_A^t)$ ;
4  $M' \leftarrow \text{Rows}(\hat{\mathbf{P}}_{S,1}^t)$  and  $\mathbf{P}_{S,1}^t \leftarrow \hat{\mathbf{P}}_{S,1}^t$ ;
5 if  $M' < M$  then
6   Generate  $\mathbf{P}_{S,\text{rand}}^t \in \mathbb{Z}_{>0}^{(M-M') \times 4}$  with entries sampled from  $\{1, \dots, N_{\text{exp}}\}$ ;
7   Restore the population  $\mathbf{P}_{S,1}^t \leftarrow \begin{bmatrix} \hat{\mathbf{P}}_{S,1}^t \\ \mathbf{P}_{S,\text{rand}}^t \end{bmatrix}$ ;
8 return  $(\mathbf{R}_1^t, \mathbf{P}_{S,1}^t)$ 

```

C Struct Population and Value Population

The struct population can be described as

$$\mathbf{P}_{S,i}^t = \begin{bmatrix} (\mathbf{n}_{1,i}^t)^\top \\ \vdots \\ (\mathbf{n}_{m,i}^t)^\top \\ \vdots \\ (\mathbf{n}_{M,i}^t)^\top \end{bmatrix} = \begin{bmatrix} N_{1,1,i}^t & N_{2,1,i}^t & N_{3,1,i}^t & N_{4,1,i}^t \\ \vdots & \vdots & \vdots & \vdots \\ N_{1,m,i}^t & N_{2,m,i}^t & N_{3,m,i}^t & N_{4,m,i}^t \\ \vdots & \vdots & \vdots & \vdots \\ N_{1,M,i}^t & N_{2,M,i}^t & N_{3,M,i}^t & N_{4,M,i}^t \end{bmatrix}. \quad (19)$$

The value population can be described as the four partitioned structural categories:

$$\mathbf{P}_{V,i}^t = [\mathbf{P}_{V_1,i}^t \quad \mathbf{P}_{V_2,i}^t \quad \mathbf{P}_{V_3,i}^t \quad \mathbf{P}_{V_4,i}^t] \in \mathbb{R}^{M \times 6N_{\text{exp}}}, \quad (20)$$

where the first block contains the candidate time variations of the first-order numerator terms:

$$\mathbf{P}_{V_1,i}^t = \begin{bmatrix} (\mathbf{v}_{1,1,i}^t)^\top \\ \vdots \\ (\mathbf{v}_{1,m,i}^t)^\top \\ \vdots \\ (\mathbf{v}_{1,M,i}^t)^\top \end{bmatrix} = \begin{bmatrix} O_{1,1,1,i}^t & O_{1,2,1,i}^t & \cdots & O_{1,N_{\text{exp}},1,i}^t \\ \vdots & \vdots & \ddots & \vdots \\ O_{1,1,m,i}^t & O_{1,2,m,i}^t & \cdots & O_{1,N_{\text{exp}},m,i}^t \\ \vdots & \vdots & \ddots & \vdots \\ O_{1,1,M,i}^t & O_{1,2,M,i}^t & \cdots & O_{1,N_{\text{exp}},M,i}^t \end{bmatrix} \in \mathbb{R}^{M \times N_{\text{exp}}}, \quad (21)$$

in which, $O_{1,q,m,i}^t$ denotes the time variation of the q -th candidate first-order numerator term in the m -th individual.

The second block contains the frequency and damping ratio pairs of the second-order numerator terms:

$$\mathbf{P}_{V_2,i}^t = \begin{bmatrix} (\mathbf{v}_{2,1,i}^t)^\top \\ \vdots \\ (\mathbf{v}_{2,m,i}^t)^\top \\ \vdots \\ (\mathbf{v}_{2,M,i}^t)^\top \end{bmatrix} = \begin{bmatrix} \omega_{2,1,1,i}^t & \xi_{2,1,1,i}^t & \cdots & \omega_{2,N_{\text{exp}},1,i}^t & \xi_{2,N_{\text{exp}},1,i}^t \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \omega_{2,1,m,i}^t & \xi_{2,1,m,i}^t & \cdots & \omega_{2,N_{\text{exp}},m,i}^t & \xi_{2,N_{\text{exp}},m,i}^t \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \omega_{2,1,M,i}^t & \xi_{2,1,M,i}^t & \cdots & \omega_{2,N_{\text{exp}},M,i}^t & \xi_{2,N_{\text{exp}},M,i}^t \end{bmatrix} \in \mathbb{R}^{M \times 2N_{\text{exp}}}, \quad (22)$$

where each pair $(\omega_{2,q,m,i}^t, \xi_{2,q,m,i}^t)$ represents the frequency and damping ratio of the q -th candidate second-order numerator term.

The third block contains the candidate time variations of the first-order denominator terms:

$$\mathbf{P}_{V_3,i}^t = \begin{bmatrix} (\mathbf{v}_{3,1,i}^t)^\top \\ \vdots \\ (\mathbf{v}_{3,m,i}^t)^\top \\ \vdots \\ (\mathbf{v}_{3,M,i}^t)^\top \end{bmatrix} = \begin{bmatrix} O_{3,1,1,i}^t & O_{3,2,1,i}^t & \cdots & O_{3,N_{\text{exp}},1,i}^t \\ \vdots & \vdots & \ddots & \vdots \\ O_{3,1,m,i}^t & O_{3,2,m,i}^t & \cdots & O_{3,N_{\text{exp}},m,i}^t \\ \vdots & \vdots & \ddots & \vdots \\ O_{3,1,M,i}^t & O_{3,2,M,i}^t & \cdots & O_{3,N_{\text{exp}},M,i}^t \end{bmatrix} \in \mathbb{R}^{M \times N_{\text{exp}}}, \quad (23)$$

in which, $O_{3,q,m,i}^t$ denotes the time constant of the q -th candidate first-order denominator term in the m -th individual.

The fourth block contains the frequency and damping ratio pairs of the second-order denominator terms:

$$\mathbf{P}_{V_4,i}^t = \begin{bmatrix} (\mathbf{v}_{4,1,i}^t)^\top \\ \vdots \\ (\mathbf{v}_{4,m,i}^t)^\top \\ \vdots \\ (\mathbf{v}_{4,M,i}^t)^\top \end{bmatrix} = \begin{bmatrix} \omega_{4,1,1,i}^t & \xi_{4,1,1,i}^t & \cdots & \omega_{4,N_{\text{exp}},1,i}^t & \xi_{4,N_{\text{exp}},1,i}^t \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \omega_{4,1,m,i}^t & \xi_{4,1,m,i}^t & \cdots & \omega_{4,N_{\text{exp}},m,i}^t & \xi_{4,N_{\text{exp}},m,i}^t \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \omega_{4,1,M,i}^t & \xi_{4,1,M,i}^t & \cdots & \omega_{4,N_{\text{exp}},M,i}^t & \xi_{4,N_{\text{exp}},M,i}^t \end{bmatrix} \in \mathbb{R}^{M \times 2N_{\text{exp}}}. \quad (24)$$

where the pair $(\omega_{4,q,m,i}^t, \xi_{4,q,m,i}^t)$ represents the frequency and damping ratio of the q -th candidate second-order denominator term.

D Prompt Template

We present the prompt template of Structure Designer in Figure 9 and the prompt template of Structure Advisor in Figure 10.

E Experimental Settings

E.1 Metrics Definition

Mean Success Rate (MSR). Let $\mathcal{P}_{\text{Pareto}}^t$ denote the final Pareto control parameter set obtained at flight stage t , and let $\widehat{\mathbf{F}}^t = [\widehat{F}_1^t, \dots, \widehat{F}_8^t]^\top$ denote the expected objective thresholds specified in Table 1. A control parameter $\mathbf{x} \in \mathcal{P}_{\text{Pareto}}^t$ is considered successful only if all stage-level objectives satisfy their expected thresholds. Its success criterion is defined as

$$\delta^t(\mathbf{x}) = \prod_{j=1}^8 \mathbb{I} \left[F_j^t(\mathbf{x}) \geq \widehat{F}_j^t \right], \quad (25)$$

where $\mathbb{I}[\cdot]$ is the indicator function. The Mean Success Rate averages the success rates over all T flight stages:

$$\text{MSR} = \frac{1}{T} \sum_{t=1}^T \frac{1}{|\mathcal{P}_{\text{Pareto}}^t|} \sum_{\mathbf{x} \in \mathcal{P}_{\text{Pareto}}^t} \delta^t(\mathbf{x}). \quad (26)$$

A larger MSR indicates that a higher proportion of the obtained Pareto control parameters simultaneously satisfy all expected flight attitude requirements across the dynamic flight process.

Mean Disparity (MD). Consistent with the above definition, let $\widehat{\mathbf{F}}^t = [\widehat{F}_1^t, \dots, \widehat{F}_8^t]^\top$ denote the expected objective thresholds for flight stage t , where all objectives and thresholds follow the unified maximization form. For a Pareto control parameter $\mathbf{x} \in \mathcal{P}_{\text{Pareto}}^t$, its

[Role]

You are the Structure Designer in LLMCtrl, an LLM agent responsible for assisting variable structure control parameter design in Dynamic Launch Vehicle Attitude Control. Your task is to identify promising candidate positions within the value population and generate a rank matrix to guide the construction of activation matrices. You do not directly optimize numerical parameter values, modify the populations, or generate complete control parameters.

[Launch Vehicle Knowledge]

The following content provides commonsense and domain knowledge related to launch vehicle attitude control: $[\mathcal{K}_V]$. Use this knowledge to understand how control parameters regulate the attitude of a launch vehicle and how different numerator and denominator terms may affect control performance.

[DLVAC Description]

The following content describes the Dynamic Launch Vehicle Attitude Control problem: $[\mathcal{D}_{\text{DLVAC}}]$. A launch vehicle flight process consists of multiple continuous flight stages. Each flight stage is represented by multiple transfer functions, and one control parameter must regulate all transfer functions within the current stage. The control parameter follows a variable structure and consists of four types of terms: first-order numerator terms, second-order numerator terms, first-order denominator terms, and second-order denominator terms. Different structures correspond to different parameter dimensions and may produce different attitude control performance.

[Tasks Description]

The following content describes the current DLVAC task and flight stage: $[\mathcal{D}_{\text{task}}]$. Use the transfer function characteristics, flight stage information, control objectives, parameter ranges, and task constraints provided in this description when evaluating candidate structure positions. The control objectives are obtained from Bode plots and jointly characterize the attitude control performance of all transfer functions in the current stage.

[Current Coevolution Populations]

The current Structure Population is: $[\mathbf{P}_{S,i}^t]$. Each row is a structure individual. Its four entries specify the required numbers of first-order numerator terms, second-order numerator terms, first-order denominator terms, and second-order denominator terms. The current Value Population is: $[\mathbf{P}_{V,i}^t]$. The value population is divided into four blocks corresponding to the four types of structure terms. Each block contains candidate positions and their current numerical values. The structure population determines how many positions should be selected from each block, while your output determines which positions should receive higher activation priority.

[Domain-Guided Structure Reasoning]

Perform structured reasoning based on the provided domain knowledge, task description, structure population, and value population:

- Understand the relationship between variable structure control parameters and launch vehicle attitude control.
- Examine the distribution of the four structure term counts in $\mathbf{P}_{S,i}^t$.
- Analyze the candidate values and positions in the four blocks of $\mathbf{P}_{V,i}^t$.
- Assess which candidate positions are more likely to form effective control parameter structures for the current flight stage.
- Assign a rank score to every candidate position.

The generated rank matrix is shared by all individuals in the current population. For each structure individual, the algorithm will select the positions with the largest scores according to the required number of terms and construct the corresponding activation matrix.

[Output Constraints]

Your output must satisfy all of the following requirements:

- Output one rank matrix denoted by $\mathbf{R}_{S,i}^t$.
- The rank matrix must contain four rows. The rows correspond, in order, to first-order numerator terms, second-order numerator terms, first-order denominator terms, and second-order denominator terms.
- Each row must contain exactly N_{exp} numerical rank scores, with one score for each candidate position.
- A larger score must indicate a higher activation priority. Scores within the same row should provide a clear ordering of candidate positions.
- Do not change $\mathbf{P}_{S,i}^t$ or $\mathbf{P}_{V,i}^t$.
- Do not directly output activation matrices, value individuals, complete control parameters, Bode plot indicators, or fitness values.
- Do not add or remove candidate positions.
- Perform the detailed reasoning internally and provide only a concise design rationale with the final result.
- Return the result using exactly the following format:
Rank_Matrix: $[[\text{score}_{1,1}, \text{score}_{1,2}, \dots, \text{score}_{1,N_{\text{exp}}}],$
 $[\text{score}_{2,1}, \text{score}_{2,2}, \dots, \text{score}_{2,N_{\text{exp}}}],$
 $[\text{score}_{3,1}, \text{score}_{3,2}, \dots, \text{score}_{3,N_{\text{exp}}}],$
 $[\text{score}_{4,1}, \text{score}_{4,2}, \dots, \text{score}_{4,N_{\text{exp}}}]$

Before returning the result, verify that the matrix contains four rows, that every row contains exactly N_{exp} scores, and no additional decision variables are included.

Figure 9: The prompt template of Structure Designer.

disparity is defined as

$$\text{Dis}^t(\mathbf{x}) = \begin{cases} \sum_{j=1}^8 \max \{ \widehat{F}_j^t - F_j^t(\mathbf{x}), 0 \}, & \delta^t(\mathbf{x}) = 0, \\ - \sum_{j=1}^8 (F_j^t(\mathbf{x}) - \widehat{F}_j^t), & \delta^t(\mathbf{x}) = 1, \end{cases} \quad (27)$$

[Role]
You are the Structure Advisor in LLMCtrl, an LLM agent responsible for transferring control parameter design knowledge across continuous flight stages. When the flight stage changes, your task is to reason over historical design trajectories and recommend a promising initial structural search state for the new flight stage. You do not perform population evolution or directly generate complete control parameters.

[Flight Stage Dynamics Knowledge]
The following content provides domain knowledge about the dynamic relationships among launch vehicle flight stages: $[K_{\text{DYN}}]$. A launch vehicle flight process should be viewed as a continuous process rather than a collection of independent control tasks. Consecutive flight stages may exhibit gradual changes in vehicle dynamics, transfer function characteristics, and control requirements. Therefore, effective structural patterns from previous stages may provide useful guidance for the new stage, but they should not be copied without considering stage changes.

[DLVAC Description]
The following content describes the Dynamic Launch Vehicle Attitude Control problem: $[D_{\text{DLVAC}}]$. Each flight stage contains multiple transfer functions that must be regulated by one variable structure control parameter. The control parameter consists of four types of terms: first-order numerator terms, second-order numerator terms, first-order denominator terms, and second-order denominator terms. Its structure determines the number of terms, while its activation pattern determines the selected candidate positions. Both the decision space and the control objectives may change across flight stages.

[Tasks Description]
The following content describes the target flight stage: $[D_{\text{task}}]$. Use the transfer function characteristics, flight stage conditions, control objectives, structural limits, and parameter ranges provided for the new stage. Pay particular attention to the differences between the target stage and the historical stages represented in the design trajectories.

[Historical Design Trajectories]
The following content contains the K most recent design trajectories: $[H_{K+1}^t]$. Each trajectory is generated after a Structure Designer interaction and contains Pareto structure designs from a previous optimization state. Each record pairs an activation matrix $A_{m,i}^u$ with its corresponding structure individual $n_{m,i}^u$.

The activation matrix records which candidate positions were selected for the four structure term types. The structure individual records how many terms were used for each type. The trajectories are temporally ordered and jointly describe how promising control parameter structures evolved across previous flight stages.

[Trajectory-Guided Structure Reasoning]
Analyze the trajectories in temporal order.
(a) Identify structure term counts that repeatedly appear among Pareto designs.
(b) Identify candidate positions that are consistently activated across trajectories.
(c) Distinguish stable structural patterns that change with the flight stage.
(d) Examine the joint relationship between structure term and activation positions.
(e) Compare historical flight conditions with the target stage and determine which patterns are transferable.
(f) Construct a promising but diverse structural initialization for the new stage:
(i) Rank Matrix Recommendation: Generate an initial rank matrix denoted by R_1^{t+1} . The rank matrix must summarize the activation priority of all candidate positions for the target stage. It contains four rows corresponding, in order, to first-order numerator terms, second-order numerator terms, first-order denominator terms, and second-order denominator terms. Each row contains N_{exp} rank scores. A larger score indicates that the position should receive a higher activation priority.
(ii) Structure Candidate Recommendation: Generate a set of initial structure candidates denoted by $\hat{P}_{S,1}^{t+1}$. Each candidate must contain four positive integers corresponding, to the numbers of first-order numerator terms, second-order numerator terms, first-order denominator terms, and second-order denominator terms. Every integer must be between 1 and N_{exp} . The number of generated candidates is denoted by M' . The candidates should represent promising structural patterns inferred from historical trajectories while maintaining sufficient structural diversity. Avoid generating duplicate candidates unless the historical evidence strongly supports repeated initialization.
Do not assume that the most recent or most frequent historical structure is automatically optimal for the target stage. Historical information should guide the initialization while preserving the ability to adapt to new flight conditions.

[Output Constraints]
Your output must satisfy all of the following requirements:
(a) Output exactly one initial rank matrix R_1^{t+1} .
(b) The rank matrix must contain four rows, each row contain N_{exp} numerical scores.
(c) Output one structure candidate set $\hat{P}_{S,1}^{t+1}$.
(d) Each structure candidate must contain exactly four integer values.
(e) Every integer in a structure candidate must lie between 1 and N_{exp} .
(f) The number of structure candidates M' must satisfy $1 \leq M' \leq M$.
(g) Do not output the value population, complete control parameters, activation matrices for individual candidates, Bode plot indicators, or fitness values.
(h) Do not copy historical candidates without considering the new flight stage.
(i) Perform the detailed reasoning internally and provide only a concise transfer rationale with the final result.
(j) Return the result using exactly the following format:
Initial_Rank_Matrix: Structure_Candidates:
[[score_{1,1}, score_{1,2}, ..., score_{1,N_{exp}}], [[N_{1,1}, N_{2,1}, N_{3,1}, N_{4,1}],
[score_{2,1}, score_{2,2}, ..., score_{2,N_{exp}}], [N_{1,2}, N_{2,2}, N_{3,2}, N_{4,2}],
[score_{3,1}, score_{3,2}, ..., score_{3,N_{exp}}], [... ],
[score_{4,1}, score_{4,2}, ..., score_{4,N_{exp}}]] [N_{1,M'}, N_{2,M'}, N_{3,M'}, N_{4,M'}]]
Before returning the result, verify the rank matrix dimensions, the number of structure candidates, the range of every structure value, and the absence of duplicated or invalid candidates.

Figure 10: The prompt template of Structure Advisor.

where $\delta^t(\mathbf{x})$ is the success indicator defined in Equation (25). For an unsuccessful solution, only the objectives that fail to reach their

expected thresholds contribute to the disparity, resulting in a positive value. For a successful solution, the disparity is the negative sum of the margins beyond all expected thresholds, such that a solution with larger performance margins obtains a smaller value.

To evaluate the entire dynamic flight process, we average the stage-level disparities over all T flight stages:

$$\text{MD} = \frac{1}{T} \sum_{t=1}^T \frac{1}{|\mathcal{P}_{\text{Pareto}}^t|} \sum_{\mathbf{x} \in \mathcal{P}_{\text{Pareto}}^t} \text{Dis}^t(\mathbf{x}). \quad (28)$$

A lower MD indicates better overall control quality. Specifically, a positive value indicates that the Pareto set still contains objective violations on average, whereas a nonpositive value indicates that the expected requirements are satisfied on average, with a more negative value representing larger performance margins beyond the expected thresholds.

E.2 Adaptation for Optimization Algorithms

Existing optimization algorithms cannot be directly applied to DLVAC tasks because the structural variability of control parameters induces changes in the dimensionality of the decision space. LLMCtrl addresses this challenge by employing LLM agents to design diverse parameter structures.

To adapt optimization algorithms with minimal modification, we introduce a population encoding that supports variable-structure optimization, as illustrated in Figure 11. Specifically, each population is divided into a structure decision component, similarly to $P_{S,i}^t$, and a parameter value component, similarly to $P_{V,i}^t$. Each entry in the structure decision component is a real value in $[0, 1]$. An entry greater than 0.5 activates the control parameter structure term at the corresponding position and associates it with its numerical value in the parameter value component. This encoding enables algorithms to optimize both parameter structures and values. However, the adapted algorithms still differ substantially from LLMCtrl. Without access to relevant domain knowledge, they cannot explicitly identify or construct appropriate control parameter structures and instead rely solely on evolutionary search to explore different structural configurations.

E.3 Control Comparison via Bode plots.

Figures 12–15 compare the control performance of LLMCtrl and the competing methods under different change frequencies using Bode plots. Each curve represents the response of an individual transfer function, from which the quantitative control indicators are extracted according to the evaluation rules defined in Appendix A.3. Such plots provide an intuitive engineering interpretation because domain experts also evaluate designed control parameters by visually examining the collective trends of the transfer function responses. Successful control is achieved when these responses are shifted toward the desired magnitude and phase regions for the corresponding flight stage.

Notably, the previously presented Bode plots focus on the high-frequency band. To provide a more comprehensive view, Figure 16 presents the full frequency band responses of LLMCtrl for Task-5 under a change frequency of 800 as an illustrative example. For the selected stage, we report the low, mid, and high frequency responses before and after control, which complements the high frequency

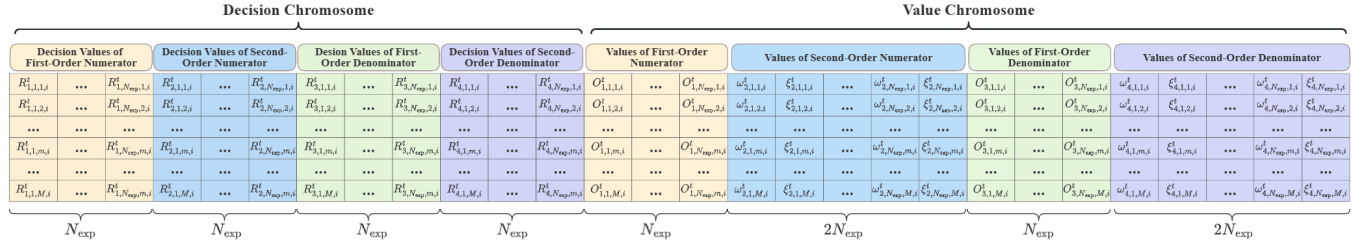
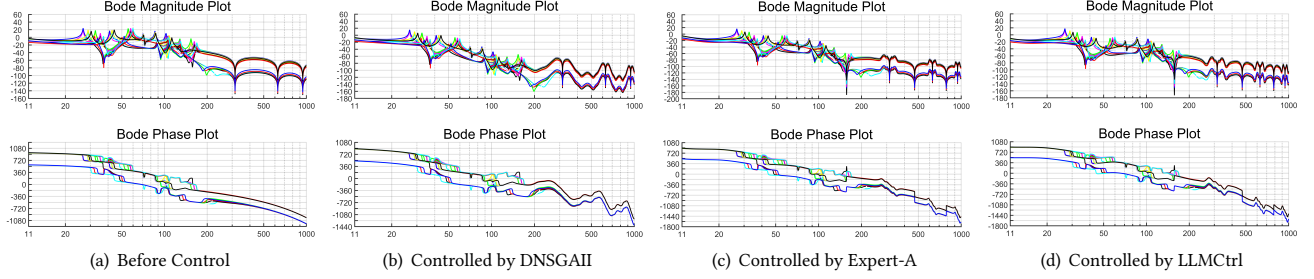
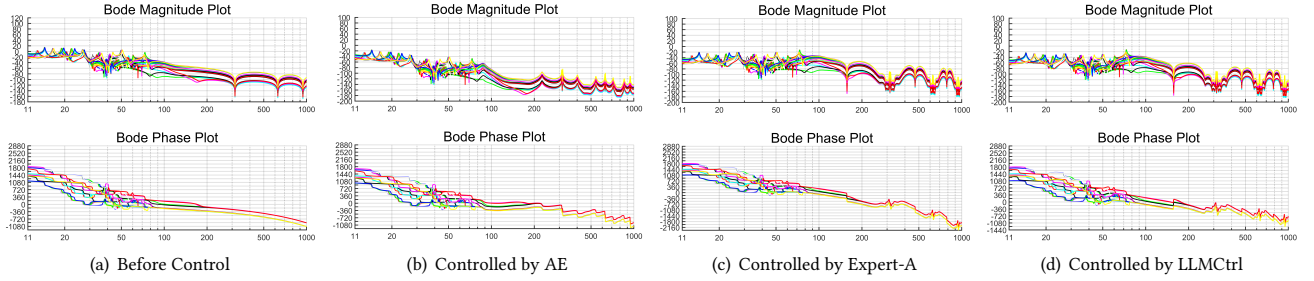
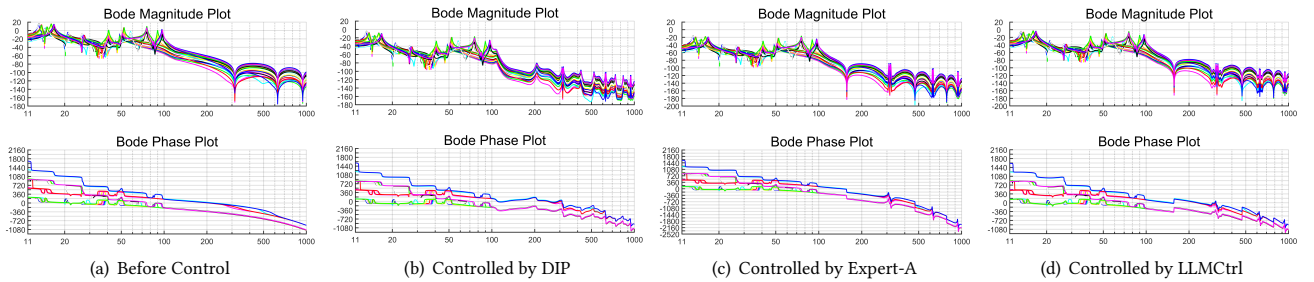


Figure 11: The population design of optimization algorithms for adaptation to DLVAC tasks.

Figure 12: Bode plots comparing LLMCtrl with competing methods on Task-1 at Stage-2 under a change frequency of 200. The axes report frequency in (rad/s), magnitude in (dB), and phase in ($^{\circ}$).Figure 13: Bode plots comparing LLMCtrl with competing methods on Task-2 at Stage-4 under a change frequency of 800. The axes report frequency in (rad/s), magnitude in (dB), and phase in ($^{\circ}$).Figure 14: Bode plots comparing LLMCtrl with competing methods on Task-3 at Stage-6 under a change frequency of 200. The axes report frequency in (rad/s), magnitude in (dB), and phase in ($^{\circ}$).

band results. These plots offer a complete view of how LLMCtrl

drives the launch vehicle to satisfy the design constraints across the entire frequency band.

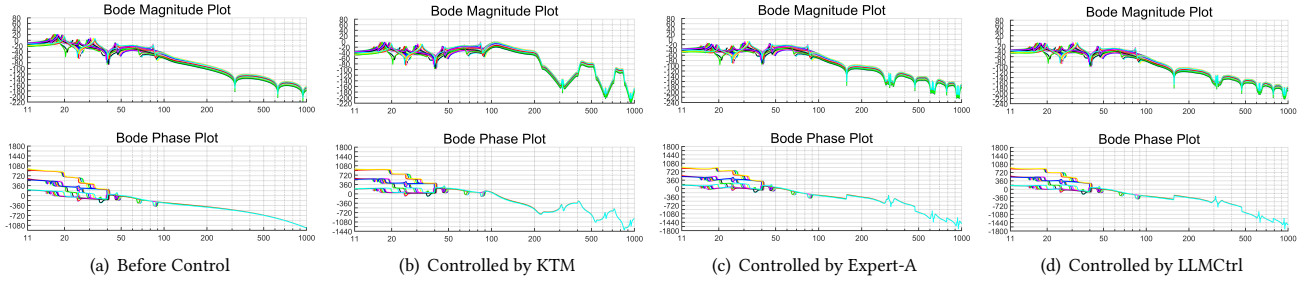


Figure 15: Bode plots comparing LLMCtrl with competing methods on Task-5 at Stage-8 under a change frequency of 800. The axes report frequency in (rad/s), magnitude in (dB), and phase in ($^{\circ}$).

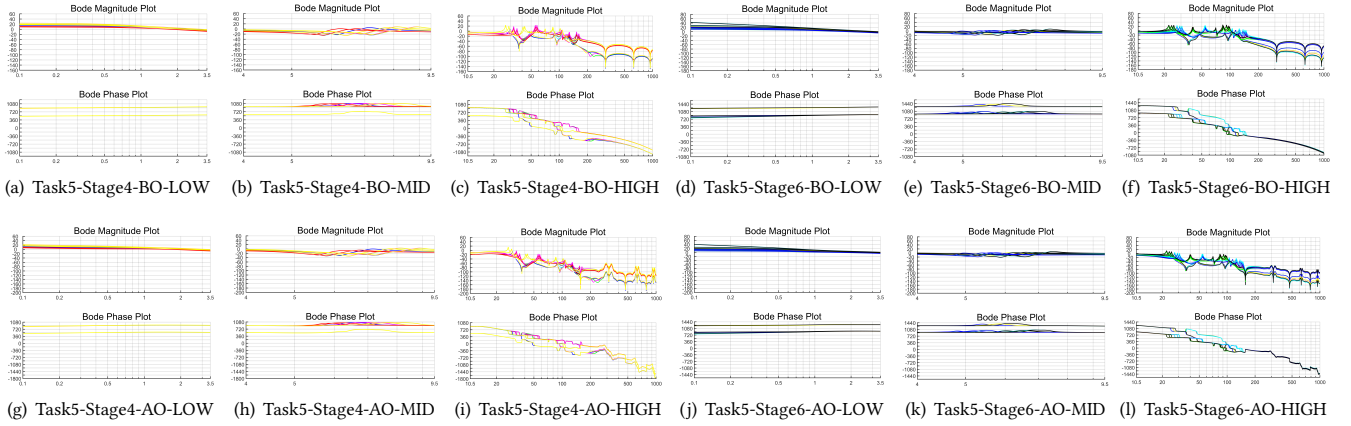


Figure 16: Full frequency band Bode plots for Task-5 controlled by LLMCtrl under a change frequency of 800. (a)–(c) show the low, mid, and high frequency responses of the launch vehicle before control in the 4th stage, and (g)–(i) show the corresponding responses after control. (d)–(f) show the low, mid, and high frequency responses before control in the 6th stage, and (j)–(l) show the corresponding responses after control.